Spacetime Thermodynamics and Singularities

by

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Abstract

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Over 60 years ago, Penrose proposed a theorem predicting the occurrence of a singularity in a generic spacetime despite the contemporary belief that a singularity is caused by a high degree of symmetry. However, the Null Energy Condition (NEC), one of the assumptions in Penrose singularity theorem, can be violated by quantum fields. Recently, Wall in 2013 improved Penrose's theorem in the semiclassical regime using spacetime thermodynamics. He replaced the NEC with the Generalized Second Law (GSL), which states that the entropy of a future causal horizon plus the entropy of matter inside the past of the horizon never decreases. This result poses a constraint on any attempt to remove singularities in quantum gravity without violating the GSL.

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Abbreviations

SEC	Strong Energy Condition
WEC	Weak Energy Condition
NEC	Null Energy Condition
OSL	Ordinary Second Law
GSL	Generalized Second Law
QFC	Quantum Focusing Conjecture

Conventions

The metric signature is $-\underbrace{++\cdots+}_{D-1 \text{ times}}$ and the units are adjusted so that $c = k_B = 1$. Additionally, the abstract index notation for tensors is used as employed in the textbook "General Relativity" by R. Wald [1].

Symbols

$\emptyset$	the empty set
$\mathrm{int}(S)$	(topological) interior of the set S
$\overline{S}$	closure of the set S
∂S	boundary of the set S
$I^+(S)/I^-(S)$	chronological future/past of the set S
$J^+(S)/J^-(S)$	causal future/past of the set S
$D^+(S)/D^-(S)$	future/past domain of dependence of the set S
$H^+(S)/H^-(S)$	future/past Cauchy horizon of the set S

Chapter 1

Introduction

1.1 What is a Singularity in Spacetime?

One year after Einstein published his theory of general relativity, the first exact solution to Einstein's field equations was found by Karl Schwarzschild [2]. This solution

$$ds^2 = -\left(1 - \frac{2Gm}{r}\right) dt^2 + \left(1 - \frac{2Gm}{r}\right)^{-1} dr^2 + r^2\, d\Omega^2 \tag{1.1}$$

which describes the spherically symmetric vacuum spacetime, has two apparent singularities: one at the radius $r = 2Gm$ and another at the origin $r = 0$. The first singularity is a coordinate singularity, which can be removed by using the Eddington-Finkelstein coordinates [3, 4]. The second singularity at $r = 0$ is a true singularity because the curvature scalar $R^{abcd}R_{abcd} = 48Gm^2/r^6$ diverges at $r = 0$, which means that the spacetime cannot be extended beyond $r = 0$. Any observer or particle that reaches $r = 0$ will end their existence there, and the laws of physics cannot predict what happens beyond that point. It was believed for a long time that such singularities occur due to the high degree of symmetry in the Schwarzschild solution. In a realistic spacetime, it was thought that matter could avoid collapsing to a single point, and thus singularities would not form. However, in 1965, Roger Penrose proved that predictability breaks down even in a generic spacetime without any symmetries, as long as there exists a trapped surface and the matter satisfies a certain condition of positivity of energy [5]. We will discuss this result in more detail in Chapter 3.

In general, we may be tempted to intuitively define a singularity as a place in spacetime near which the curvature is unbounded and hence, by Einstein's equation, the energy density is enormously large. Singularities defined in this way are called *curvature singularities*. To be more precise, we must clarify the meaning of "near" and "unbounded". As a singularity point must be excluded from the spacetime, any curve approaching it cannot be extended beyond the singularity. Hence, we can take the limit of a suitable quantity measuring the curvature along any such inextendible curves. As the value of the curvature tensor R_{abcd} depends on the coordinate system used to calculate it, the first consideration is to use a coordinate-independent scalar polynomial formed out of g_{ab}, ϵ_{abcd}, and R_{abcd}. If this scalar polynomial blows up as we move along an inextendible curve, this means that we encounter a *scalar polynomial curvature singularity* or *s.p. curvature singularity*. As we have seen above, the scalar $R_{abcd}R^{abcd}$ is one possible quantity to examine. However, in certain spacetimes such as plane wave solutions, all scalar polynomials are zero even though the curvature tensor is non-zero [6]. An alternative option is to consider an orthonormal basis E^a_μ parallelly transported along an inextendible curve and calculate components $R_{\mu\nu\rho\sigma} = R_{abcd}E^a_\mu E^b_\nu E^c_\rho E^d_\sigma$ of the curvature tensor in this basis. Since this basis always behaves nicely, if at least one component $R_{\mu\nu\rho\sigma}$ diverges along the curve, then a *parallelly propagated curvature singularity* or a *p.p. curvature singularity* occurs. Clearly, a p.p. curvature singularity is also of the s.p. type because scalar polynomials in the curvature blows up when a component $R_{\mu\nu\rho\sigma}$ diverges.

In the above discussion, we have been rather vague about inextendible curves used in the definition of curvature singularities. Beside the possibility of running into a singularity, an inextendible curve can either run off to infinity or go around in a loop. The "distance" as we travel along such curve increases unboundedly in both of these cases, while the distance remains finite as one runs into a singularity. For geodesics, an affine parameter can be used to quantify the distance since it is unique to the curve up to an affine transformation. If an affine parameter of an inextendible geodesic has either an upper bound or lower bound, then the geodesic is *incomplete*. A spacetime with at least one incomplete geodesic is said to be *geodesically incomplete*. We may expect that an incomplete geodesic always leads us to a curvature singularity, either s.p. or p.p. However, in an example such as the Taub-NUT spacetime, incomplete timelike geodesics exist but scalar polynomials in the curvature and curvature components in a parallelly transported basis along these curves remain finite along them [6]. Nevertheless, a timelike or null geodesically incomplete spacetime is physically pathological because a freely falling observer or photon cannot continue their worldline past a certain time (or affine parameter in the case of photon). Therefore, we shall consider a timelike or null geodesically incomplete spacetime as a singular spacetime. This is a powerful notion since we can examine the global structure of spacetime to prove that it is singular under certain physically plausible conditions. Notable examples of such a result are Penrose's theorem in 1965 for null geodesic incompleteness [5], Hawking's theorem in 1967 for timelike geodesic incompleteness [7], and Penrose-Hawking singularity theorem in 1970 [8].

1.2 Spacetime Thermodynamics

It has been well-known that all thermodynamic systems composed of any form of matter obey the laws of thermodynamics. However, it was not evident that spacetime itself could be also treated as thermodynamic system because it had been traditionally considered as a mere background. Nonetheless, since Einstein made it dynamical in general relativity, this point of view about spacetime ought to be changed. Inspired by Hawking's area theorem in 1971 [9] which the area of a black hole is non-decreasing, Bekenstein in 1972 proposed the idea of black hole entropy which is proportional to black hole area [10]. In 1973, the four laws of black hole mechanics put forward by Bardeen, Carter, and Hawking given some energy conditions and the cosmic censorship conjecture [11] resemble the four laws of thermodynamics (see Table 1.1). In 1974, Hawking showed that a Schwarzschild black hole can radiate and gave an exact formula for the temperature of the black hole [12]

$$T_{\mathrm{H}} = \frac{\hbar\kappa}{2\pi}, \tag{1.2}$$

where κ is the surface gravity of the black hole. By comparing with the first law (see Table 1.1), we obtain the formula of the black hole entropy or *Bekenstein-Hawking entropy*

$$S_{\mathrm{BH}} = \frac{A}{4G\hbar}, \tag{1.3}$$

where A is the surface area of the black hole. Furthermore, in the same paper that introduced the black hole entropy, Bekenstein also proposed the Generalized Second Law (GSL) which states that the *generalized entropy*

$$S_{\mathrm{gen}} = S_{\mathrm{BH}} + S_{\mathrm{out}} = \frac{A}{4G\hbar} + S_{\mathrm{out}}, \tag{1.4}$$

where S_{out} is the entropy of all matter outside the black hole, is always non-decreasing. As the Bekenstein-Hawking term decreases in the case of Hawking radiation and the S_{out} decreases when matter falls into the black hole, this conjecture has been used to preserve the second law of thermodynamics. Another interpretation is that the information of matter inside the black hole is actually encoded on the event horizon, although it seems to be hidden from an observer outside. Moreover, Gibbons and Hawking in 1977 showed that the laws of black hole thermodynamics can be extended to de Sitter horizons as particle creation also occurs near these horizons [13]. More generally, Jacobson and Parentani argued that these laws are applicable to any causal horizons including Rindler horizons [14]. In 2013, Wall proved a generalization of Penrose's original singularity theorem using GSL to replace the Null Energy Condition (NEC). We will come back to this point in Chapter 4.

TABLE 1.1: Black Holes and Thermodynamics (excerpted from [1])

Law	**Thermodynamics**	**Black Holes**
Zeroth	T constant throughout body in thermal equilibrium	κ constant over horizon of stationary black hole
First	$\mathrm{d}E = T\ \mathrm{d}S + \text{work terms}$	$\mathrm{d}M = \frac{1}{8\pi G}\kappa\ \mathrm{d}A + \Omega_H\ \mathrm{d}J$
Second	$\delta S \geq 0$ in any process	$\delta A \geq 0$ in any process
Third	Impossible to achieve $T = 0$ by a physical process	Impossible to achieve $\kappa = 0$ by a physical process

1.3 Organization of the Thesis

After this chapter, this thesis continues with four more chapters. Chapter 2 presents some important concepts of the causal structure of a spacetime that are needed for the singularity theorems in subsequent chapters. The notations and most of the details are borrowed from Chapter 8 of Wald's well-known textbook on general relativity by [1]. Moreover, some important results taken from Hawking & Ellis (1973) [6] are also added to the discussion. In Chapter 3, we discuss how energy and matter affect how a family of light rays behaves. At the end of the chapter, we present the first singularity theorem ever proposed by Penrose in 1965 [5] along with its modern proof. The derivations and discussions in Chapter 3 are influenced by Chapter 9 of Wald's textbook [1] and the lecture notes on black holes by Reall [15]. In Chapter 4, we present a recent development due to Wall who used the GSL to prove a quantum singularity theorem [16]. In addition, some results in Wall's original paper are written in terms of quantum expansion introduced by Bousso 3 years later [17]. This usage of the quantum expansion in this context can also be found in [18]. Finally, Chapter 5 discusses some advancements after Wall's discovery up until nowadays.

Chapter 2

Causal Structure of Spacetime

2.1 Pasts and Futures

To define pasts and futures of a set of events, we need to continuously classify each half of the light cone at every point in a spacetime as either past or future light cone. Thus, we define a spacetime (M, g_{ab}) to be *time-orientable* if there exists a continuous non-vanishing vector field t^a that is timelike everywhere on M. Since this vector field points to the future, every vector in the same half of the light cone as t^a also points to the future. Therefore, we can divide the set of all non-spacelike (either timelike or null) vectors at every point $p \in M$ into two disjoint classes: those *future-directed* vectors v^a satisfying $g_{ab}v^a t^b < 0$ and those *past-directed* vectors v^a satisfying $g_{ab}v^a t^b > 0$. We do not need to worry about the case $g_{ab}v^a t^b = 0$ since this identity implies that v^a is a spacelike vector.

From now on, we assume that (M, g_{ab}) is a time-orientable spacetime. With this assumption, a piecewise C^3 curve γ is defined as a *future-directed timelike curve* if its tangent vector is timelike and future-directed everywhere on the curve. If the tangent vector of γ is non-spacelike instead, then γ is a *future-directed causal curve*. Similarly, *past-directed timelike curves* and *past-directed causal curves* can be defined in the same way by replacing "future-directed" with "past-directed" in the aforementioned definitions.

Fix a point $p \in M$. Then, the *chronological future* $I^+(p)$ of the event p is defined as the set of events that can be reached by a future-directed timelike curve starting from p, i.e. $q \in I^+(p)$ if there exists a future-directed timelike curve $\gamma : [0, 1] \to M$ such that $\gamma(0) = p$ and $\gamma(1) = q$. If we replace "timelike curve" with "causal curve", then we obtain the *causal future* $J^+(p)$ of the event p, which contains $I^+(p)$ by the definition. These definitions also have their past counterparts by replacing "future" by "past" and "+" by "−". To avoid repetition, we will discuss definitions and results only for future counterparts. Furthermore, let S be a subset of M. Then, the chronological future $I^+(S)$ and the causal future $J^+(S)$ of S are defined as

$$I^+(S) = \bigcup_{p\in S} I^+(p), \qquad J^+(S) = \bigcup_{p\in S} J^+(p). \tag{2.1}$$

We now discuss some properties of these sets. Since the zero vector is not timelike but causal, a curve γ defined by $\gamma(t) = p$ for all t is not a timelike curve but a causal curve. Hence, $p \notin I^+(p)$ and $p \in J^+(p)$. For an arbitrary subset S, we still have $S \subset J^+(S)$, but it is not always true that $S \cap I^+(S) = \emptyset$. A set $S \subset M$ is said to be *achronal* if $I^+(S) \cap S = \emptyset$, i.e. there do not exist $p, q \in S$ such that $q \in I^+(p)$. Additionally, the chronological future $I^+(S)$ of any subset S is open because for any $p \in S$ and $q \in I^+(p)$, there exists a future-directed timelike curve connecting p and any point in a sufficiently small neighborhood of q. Also, though $J^+(S)$ is closed in the Minkowski space, $J^+(S)$ is not closed in a general spacetime. Indeed, Figure 2.1 shows a simple counterexample. Nevertheless, $\overline{J^+(S)} = \overline{I^+(S)}$ with the boundary $\partial J^+(S) = \partial I^+(S)$ generated by null geodesics because any two points connected by a causal curve that is not a null geodesics can also be joined by a timelike curve (see Proposition 4.5.10 in [6]). In addition, the boundary of $I^+(S)$ or $J^+(S)$ is an achronal hypersurface, whose geometry will be discussed extensively in Chapter 3.

Proposition 2.1. *Let S be a subset of M. Then, the set $\partial I^+(S)$, if non-empty, is a closed, embedded, achronal $(D-1)$-dimensional C^0-submanifold of M.*

Proof. See Proposition 6.3.1 in [6]. □

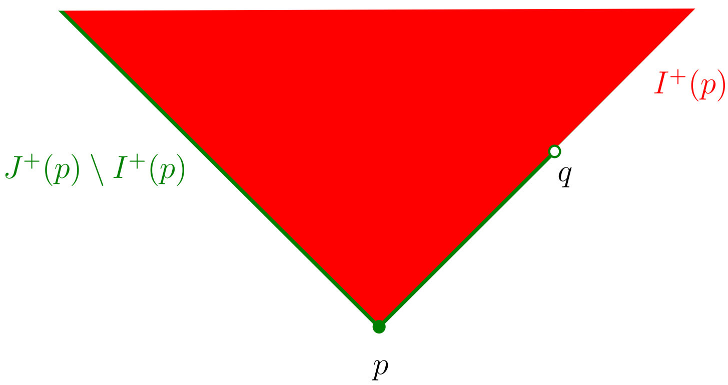

FIGURE 2.1: In a Minkowski space with the point q removed, the causal future $J^+(p)$ of the point p is not closed.

The condition of differentiability on timelike or causal curves is too restrictive if we want to extend the curves. For that purpose, it is necessary to extend the definitions of timelike and causal curves from piecewise differentiable curves to continuous curves. Even though a continuous curve may not have a well-defined tangent vector, we can still determine whether it is timelike or causal by locally approximate it by a differentiable curve. More precisely, a continuous curve $\gamma : I \to M$ is said to be a future-directed timelike curve if for every $t \in I$, there exists a convex normal neighborhood O of $\gamma(t_0)$ such that if $\gamma(t_1), \gamma(t_2) \in O$ with $t_1, t_2 \in I$ and $t_1 < t_2$, then there exists a differentiable future-directed timelike curve from $\gamma(t_1)$ to $\gamma(t_2)$. A continuous future-directed causal curve can be defined similarly by replacing "timelike" with "causal" in the aforementioned definition.

Now, consider a continuous curve $\gamma : I \to M$. The following definitions allow us to determine whether the curve γ is extendible or not. A point $p \in M$ is said to be a *future endpoint* of the curve γ if for every neighborhood O of p, there exists $t_0 \in I$ such that $\gamma(t) \in O$ for all $t > t_0$. A curve γ without any future endpoint is said to be *future inextendible*. Similarly, a point $p \in M$ is said to be a *past endpoint* of the curve γ if for every neighborhood O of p, there exists a number $t_0 \in I$ such that $\gamma(t) \in O$ for all $t \in I$ satisfying $t < t_0$. A *past inextendible* curve is then a curve without any past endpoint. A curve that is inextendible in either direction can do one of the followings: running off to infinity, circling around in a loop, and running into a singularity point. The last possibility can be distinguished from the other two by considering the distance as one travels along the curve. For a geodesic, we can consider an affine length from a fixed point on the geodesic. If the geodesic is future inextendible and has an upper bound on its affine parameter, then it is *incomplete* in the future direction. We will discuss the conditions for the existence of an incomplete null geodesic in Chapter 3 and 4.

2.2 Causality Conditions

Even though every spacetime is locally Minkowski, the global topology of some spacetimes can allow the existence of a closed timelike curve. Such a spacetime violates the *chronology condition* and allows an observer to travel back in time. It is generally believed that such a spacetime is not physically realistic. In fact, we believe that every realistic spacetime (M, g_{ab}) satisfies a stronger condition which is the *causality condition*, i.e. no closed causal curves exist in M. That is, it is not possible to send a signal to the past in a causal spacetime. Even if there is no closed causal curve, it may happen that a causal curve can approach arbitrary close to its starting point, which means that the spacetime is on the verge of violating causality. Hence, it is physically reasonable to assume a spacetime (M, g_{ab}) to be *strongly causal*, i.e. for all points $p \in M$ and every neighborhood O of p, there exists another neighborhood V of p contained in O such that no causal curve intersects V more than once. However, even this condition is not enough to rule out all spacetimes which almost violate the causality condition. There are spacetimes where a modification of the metric g_{ab} can allow a closed timelike curve to exist. Specifically, at some points, we can choose some timelike vectors ξ^a and redefine the metric at those points as follows:

$$\hat{g}_{ab} = g_{ab} - \xi_a \xi_b. \tag{2.2}$$

All non-spacelike vectors v^a with respect to g_{ab} are timelike with respect to $\hat{g}_{ab}$ because $g_{ab}v^av^b \leq 0$ and $\xi_a v^a \neq 0$ implies

$$\hat{g}_{ab}v^av^b = g_{ab}v^av^b - (\xi_a v^a)^2 < 0. \tag{2.3}$$

Therefore, this modification of the metric expands the interior of the light cone, which makes it easier for a closed timelike curve to form in a spacetime that is on the verge of producing causality condition. We might want the causal structure of a spacetime to be stable against such a modification by requiring it to be *stably causal*, i.e. there exists a continuous non-vanishing timelike vector field ξ^a such that the spacetime $(M, \hat{g}_{ab})$, where the metric $\hat{g}_{ab}$ is given by (2.2), does not possess any closed timelike curves. In such a spacetime, there exists a global time function t which increases along every future-directed causal curve.

Theorem 2.2. *A spacetime (M, g_{ab}) is stably causal if and only if there exists a differentiable function t on M such that $\nabla^a t$ is a past directed timelike vector field.*

Proof. See Theorem 8.2.2 in [1] and Proposition 6.4.9 in [6]. □

Using this theorem, it is possible to show that stable causality implies strong causality (see the corollary after Theorem 8.2.2 in [1]).

2.3 Domains of Dependence and Global Hyperbolicity

In Newtonian mechanics, it is possible to specify the initial condition at a present time and then determine the past and the future everywhere in spacetime. However, the theory of relativity forbids material particles and radiation to exceed the speed of light. Therefore, an initial condition on a subset S can only influence a certain set of events. We assume that S is closed because an initial condition on an arbitrary set S can be extended to its closure $\overline{S}$ by the continuity. In addition, to prevent redundancy in the information specified on S, S is required to be achronal, so that any two events on S are independent of each other. Moreover, we may want the set S to be "wide" enough so that the spacetime region influenced by S is the largest. For a closed and achronal set S, the *edge* of S, denoted by $\mathrm{edge}\,(S)$, is defined as the set of points $p \in S$ such that every neighborhood O of p contains a point $q \in I^+(p)$, a point $r \in I^-(p)$, and a timelike curve γ from r to q that does not intersect S. Then, a closed achronal set without edge is what we are looking for. Using the same proof as in Proposition 2.1, we have the following result.

Proposition 2.3. *A non-empty closed achronal set S with* $\mathrm{edge}\,(S) = \emptyset$ *is a three-dimensional, embedded, C^0-submanifold of M.*

If a closed achronal set S without edge is a spacelike hypersurface which no causal curve intersect more than once, then it is a *partial Cauchy surface.* This surface can be considered as a moment of time.

The *future domain of dependence* $D^+(S)$ of S is defined as the set of points $p \in M$ such that every past inextendible causal curve through p intersects S. The *past domain of dependence* $D^-(S)$ of S can be defined similarly by replacing "past inextendible causal curve" by "future inextendible causal curve". The *domain of dependence* $D(S)$ of S is then defined as

$$D(S) = D^+(S) \cup D^-(S). \tag{2.4}$$

With this definition, the information at every point in $D(S)$ can be traced back to S. The boundary of this region can be defined as

$$H^+(S) = \overline{D^+(S)} \setminus I^-[D^+(S)], \qquad H^-(S) = \overline{D^-(S)} \setminus I^+[D^-(S)], \tag{2.5}$$

which are called the future and past Cauchy horizons, respectively. If a closed achronal set S has no edge, then the Cauchy horizons $H^+(S)$ and $H^-(S)$ are null hypersurfaces, i.e. a submanifold generated by null geodesics, like the boundary $\partial I^+(S)$ of the future of S (see Theorem 8.3.5 in [1]).

A partial Cauchy surface S for which $M = D(S)$ is called a full Cauchy surface or simply *Cauchy surface.* The existence of a Cauchy surface means that spacetime is predictable since all information needed to determine the past and future is contained solely on the surface S. In this case, $H^+(S)$ and $H^-(S)$ are empty because the manifold M has no boundary. In addition, a spacetime possessing a Cauchy surface actually satisfies the strongest causality condition.

A spacetime (M, g_{ab}) is said to be *globally hyperbolic* if it satisfies the following two conditions:

(i) Strong causality holds on M.

(ii) For every pair of points $p, q \in M$, the set $J^+(p) \cap J^-(q)$ is compact.

Strong causality is traditionally assumed in the definition of global hyperbolicity [1, 6]. However, it is recently shown that just causality is sufficient.

Theorem 2.4. *Assume that a spacetime* (M, g_{ab}) *satisfies the condition that* $J^+(p) \cap J^-(q)$ *is compact for all* $p, q \in M$*. Then, the causality condition and strong causality condition are equivalent.*

Proof. See Theorem 3.2 of [19]. □

It is easy to see that a spacetime having a Cauchy surface S satisfies the chronology condition. Indeed, a closed timelike curve cannot intersect S because S is achronal.

As a result, every point on the closed timelike curve is outside the domain of dependence of S. In fact, the existence of a Cauchy surface implies the global hyperbolicity.

Proposition 2.5. *If a spacetime (M, g_{ab}) admits a Cauchy surface S, then it is globally hyperbolic.*

Proof. See Proposition 6.6.3 in [6], Lemma 8.3.8 and Theorem 8.3.10 in [1]. □

Conversely, it is possible to show that a globally hyperbolic spacetime is stably causal. Then, Theorem 2.2 states that a global time function t on M exists. The surfaces of constant t turn out to be Cauchy surfaces.

Proposition 2.6. *Let (M, g_{ab}) be a globally hyperbolic spacetime. Then, M is homeomorphic to $\mathbb{R} \times S$, where S is a three-dimensional manifold, and $\{t\} \times S$ is a Cauchy surface for each $t \in \mathbb{R}$.*

Proof. See Proposition 6.6.8 in [6] and Theorem 8.3.14 in [1]. □

Therefore, the existence of a Cauchy surface is equivalent to the global hyperbolicity. It is natural to believe that any physically realistic spacetime is globally hyperbolic, so that it is possible for us to use our physical theories to predict events at any place in the spacetime. However, some known solutions of the Einstein equations fail to be globally hyperbolic such as the analytic extension of Kerr solutions which are not super-extremal [20], plane wave solutions [21]. Therefore, it is reasonable to drop the global hyperbolicity to have a general result such as the Penrose-Hawking singularity theorem in 1970 [8]. Nevertheless, we still assume the global hyperbolicity in the two central singularity theorems presented in this thesis.

Chapter 3

Classical Singularity Theorem

3.1 Null Geodesics Congruences

Let $O \subset M$ be an open subset. A *congruence* in O is a family of curves such that exactly one curve of the family passes through each point in O. The congruence is smooth if it is generated by a smooth vector field in O. In this section, we consider a smooth congruence of null geodesics, which is generated by a null vector field k^a, i.e. $g_{ab}k^a k^b = 0$. Assume that the parameter λ along each geodesic is an affine parameter, i.e. the vector field k^a satisfies the geodesic equation

$$k^a \nabla_a k^b = 0. \tag{3.1}$$

Consider a smooth one-parameter subfamily of geodesics γ_s in the congruence, i.e. the map $(\lambda, s) \mapsto \gamma_s(\lambda)$ is smooth and one-to-one with a smooth inverse. Then, this subfamily generates a smooth two-dimensional submanifold. The two parameters λ and s constitute a coordinate system (λ, s) on this manifold, which generates two coordinate basis vectors $k^a = (\partial/\partial\lambda)^a$ and $\eta^a = (\partial/\partial s)^a$. The vector η^a along a geodesic γ_s is called the *deviation vector* since it represents an infinitesimal displacement from γ_s to a nearby geodesic in the subfamily. Since k^a and η^a are coordinate basis vectors, they commute, i.e.

$$\mathcal{L}_k \eta^a = [k, \eta]^a = k^b \nabla_b \eta^a - \eta^b \nabla_b k^a = 0, \tag{3.2}$$

which implies that

$$k^b \nabla_b \eta^a = \eta^b \nabla_b k^a = B^a{}_b \eta^b, \tag{3.3}$$

where the tensor field B_{ab} is defined as

$$B_{ab} = \nabla_b k_a. \tag{3.4}$$

This tensor satisfies two important properties:

$$B_{ab}k^b = k^b\nabla_b k_a = 0 \tag{3.5}$$

due to the geodesic equation (3.1), and

$$B_{ab}k^a = k^a\nabla_b k_a = \frac{1}{2}\nabla_b(k_a k^a) = 0, \tag{3.6}$$

due to the fact that k^a is a null vector. Hence, ${B^a}_b v^b$ and ${B_b}^a v^b$ for any vector v^a are both orthogonal to k^a. Looking at (3.3), we can see that the tensor B_{ab} describes the rate of change of η^a and also measures the failure of η^a to be parallel transported along the geodesics. Hence, it contains information about how the geodesics in the congruence are moving relative to each other. Despite the failure of η^a to be parallel transported, the product $k_a\eta^a$ is constant along each geodesic in the congruence since

$$k^b\nabla_b(k_a\eta^a) = k_a k^b\nabla_b\eta^a + k^b\eta^a\nabla_b k_a = k_a\eta^b\nabla_b k^a + 0 = \frac{1}{2}\eta^b\nabla_b(k_a k^a) = 0, \tag{3.7}$$

where we have used (3.3) and (3.1) in the second equality, and the fact that k^a is a null vector in the last equality.

For the next step, notice that the deviation vector η^a is non-unique because the affine parameter is unique only up to an affine transformation $\lambda \to a\lambda + b$. Indeed, consider a new affine parameter $\lambda_1 = a(s)\lambda + b(s)$ for each geodesic γ_s in the subfamily. While the tangent vector k^a is rescaled with a factor $1/a(s)$, which does not change the inner product $k^a k_a = 0$, the new deviation vector can be expressed as

$${\eta_1}^a = \left(\frac{\partial}{\partial s}\bigg|_{\lambda_1}\right)^a = \left(\frac{\partial}{\partial s}\bigg|_{\lambda}\right)^a + \frac{\partial\lambda}{\partial s}\bigg|_{\lambda_1}\left(\frac{\partial}{\partial\lambda}\bigg|_{s}\right)^a = \eta^a + c(\lambda, s)k^a, \tag{3.8}$$

where $c(\lambda, s) = -\big(a'(s)\lambda + b'(s)\big)/a(s)$. Hence, the deviation vector η^a is determined only up to an addition of a multiple of k^a. In the cases of timelike and spacelike geodesic congruences ($k^a k_a \neq 0$), we can use this freedom to make η^a orthogonal to k^a. However, in the case of null geodesic congruences, the inner product is always invariant, i.e. $\eta^a k_a = {\eta_1}^a k_a$. To fix a "gauge condition" on η^a, consider a smooth spacelike hypersurface Σ which intersects each geodesic in the congruence O exactly once. On $O\cap\Sigma$, we pick a smooth null vector field l^a such that $l^a k_a = -1$ everywhere on $O \cap \Sigma$. This is always possible since we can express l^a as a linear combination of a timelike normal vector field and a spacelike tangent vector field on $O \cap \Sigma$. The two conditions $l^a l_a = 0$ and $l^a k_a = -1$ then determine the two coefficients in the combination. In addition, we extend l^a to the whole congruence O by parallel transporting it along each geodesic, i.e. $k^b\nabla_b l^a = 0$. With this construction, both $l^a l_a$ and $k^a l_a$ are constant everywhere on the congruence. That is, we construct a null vector field l^a in the congruence, which satisfies

$$l^a l_a = 0, \quad k^a l_a = -1, \quad k^b \nabla_b l^a = 0. \tag{3.9}$$

Now, we can decompose the deviation vector η^a as

$$\eta^a = \alpha k^a + \beta l^a + \hat{\eta}^a, \tag{3.10}$$

where $\alpha = -\eta^a l_a$, $\beta = -\eta^a k_a$, and $\hat{\eta}^a$ is a vector orthogonal to both k^a and l^a. Then, we can use the freedom in choosing the affine parameter λ to set $\alpha = 0$. Hence, by choosing l^a and considering only deviation vectors η^a orthogonal to l^a, we eliminate the ambiguity in η^a. In addition, the orthogonality between $\hat{\eta}^a$ and k^a implies that $\hat{\eta}^a$ is spacelike or a multiple of k^a or zero. Furthermore, the orthogonality between $\hat{\eta}^a$ and l^a ensures that $\hat{\eta}^a$ is either spacelike or zero. We are only interested in how $\hat{\eta}^a$ changes along each geodesic because the component βl^a is parallel transported along the geodesic, which indicates that

$$k^b \nabla_b \eta^a = k^b \nabla_b \hat{\eta}^a. \tag{3.11}$$

Physically, $\hat{\eta}^a$ represents the purely spatial part of the infinitesimal separation between geodesics in the congruence. Before moving on, let us introduce the $(D-2)$-dimensional subspace $T_\perp$ at each point in the congruence O defined as

$$T_\perp = \{v^a \mid v^a k_a = v^a l_a = 0\}. \tag{3.12}$$

Then, let us also define the symmetric tensor

$$h_{ab} = g_{ab} + k_a l_b + l_a k_b, \tag{3.13}$$

which serves as both a projection onto $T_\perp$ and a metric on $T_\perp$. That is, for any vector $v^a \in T_\perp$, the tensor h_{ab} satisfies

$$h^a{}_b k^b = h^a{}_b l^b = 0, \tag{3.14}$$

$$h^a{}_b v^b = v^a, \tag{3.15}$$

$$h_{ab} v^a = g_{ab} v^a, \tag{3.16}$$

$$h^a{}_c h^c{}_b = h^a{}_b. \tag{3.17}$$

By definition, it is clear that $\hat{\eta}^a \in T_\perp$. From (3.14) and (3.15), we have $h^a{}_b \eta^b = h^a{}_b \hat{\eta}^b = \hat{\eta}^a$. Furthermore, since k^a and l^a are parallel transported along the geodesics, h_{ab} is also parallel transported, i.e.

$$k^c \nabla_c h_{ab} = 0. \tag{3.18}$$

Then, the evolution of $\hat{\eta}^a$ along each geodesic is governed by the equation

$$k^b \nabla_b \hat{\eta}^a = k^b \nabla_b (h^a{}_c \eta^c) = h^a{}_c k^b \nabla_b \eta^c = h^a{}_c B^c{}_b \eta^b. \tag{3.19}$$

Later, we often consider a family of null geodesics starting from spatially-separated sources, i.e. we are interested only in the case $\beta = 0$ or $\eta^a \in T_\perp$. In this case, we have $\eta^b = \hat{\eta}^b$ and

$$k^b \nabla_b \hat{\eta}^a = h^a{}_c B^c{}_b \hat{\eta}^b = h^a{}_c B^c{}_b h^b{}_d \hat{\eta}^d = \hat{B}^a{}_b \hat{\eta}^b, \tag{3.20}$$

where

$$\hat{B}_{ab} = h_a{}^c h_b{}^d B_{cd} = h_a{}^c h_b{}^d \nabla_d k_c. \tag{3.21}$$

This shows that only the projected tensor $\hat{B}^a{}_b$ affects the evolution of the spacelike deviation vector η^a. Compared to B_{ab}, the projected tensor $\hat{B}_{ab}$ has two more properties in addition to (3.5) and (3.6):

$$\hat{B}_{ab} k^b = \hat{B}_{ba} k^b = \hat{B}_{ab} l^b = \hat{B}_{ba} l^b = 0, \tag{3.22}$$

which follows from (3.14). To understand the geometrical effects of $\hat{B}^a{}_b$, we decompose it into its trace, symmetric traceless, and antisymmetric parts:

$$\hat{B}_{ab} = \frac{1}{D-2} \theta h_{ab} + \hat{\sigma}_{ab} + \hat{\omega}_{ab}, \tag{3.23}$$

where the *expansion* θ, *shear* $\hat{\sigma}_{ab}$, and *rotation* $\hat{\omega}_{ab}$ are defined respectively as

$$\theta = h^{ab} \hat{B}_{ab}, \tag{3.24}$$

$$\hat{\sigma}_{ab} = \hat{B}_{(ab)} - \frac{1}{D-2} \theta h_{ab}, \tag{3.25}$$

$$\hat{\omega}_{ab} = \hat{B}_{[ab]}. \tag{3.26}$$

The factor $1/(D-2)$ in front of θ in (3.23) arises because h_{ab} acts as a positive definite metric on the $(D-2)$-dimensional subspace $T_\perp$ (all vectors in $T_\perp$ are spacelike). More explicitly, we have $h^{ab} h_{ab} = h^a{}_a = \delta^a{}_a + k^a l_a + l^a k_a = D - 1 - 1 = D - 2$. In addition, it is easy to see that both shear $\hat{\sigma}_{ab}$ and rotation $\hat{\omega}_{ab}$ are traceless, i.e. $\hat{\sigma}_{ab} h^{ab} = \hat{\omega}_{ab} h^{ab} = 0$, by using their definitions. To explain why only expansion does not have a hat, notice that it is independent of the choice of l^a since

$$\theta = h^{ab} \hat{B}_{ab} = h^{ab} h_{ac} h^d{}_b B^c{}_d = h^d{}_c B^c{}_d = \delta^d{}_c B^c{}_d + k^c l_d B^d{}_c + l^c k_d B^d{}_c = B^d{}_d = \nabla_d k^d, \tag{3.27}$$

where we have used (3.17), (3.5), and (3.6) in this derivation. Furthermore, the evolution of $\hat{B}_{ab}$ is governed by the equation

$$\begin{aligned} k^c\nabla_c\hat{B}_{ab} &= k^c\nabla_c\left(h_a{}^e h_b{}^d\nabla_d k_e\right) = k^c h_a{}^e h_b{}^d\nabla_c\nabla_d k_e \\ &= k^c h_a{}^e h_b{}^d\left(\nabla_d\nabla_c k_e + R_{cdef}k^f\right) \\ &= h_a{}^e h_b{}^d\left[\nabla_d\left(k^c\nabla_c k_e\right) - \nabla_d k^c\nabla_c k_e - R_{dcef}k^c k^f\right] \\ &= -\hat{B}_{ac}\hat{B}^c{}_b - h_a{}^e h_b{}^d R_{dcef}k^c k^f, \end{aligned} \tag{3.28}$$

where we have used (3.18) in the second and fourth equalities, and the geodesic equation (3.1) in the last equality. Taking the trace of this equation, we obtain the following important equation

$$\begin{aligned} k^c\nabla_c\theta &= -h^{ab}\hat{B}_{ac}\hat{B}^c{}_b - h^{ab}R_{acbd}k^c k^d, \\ &= -\hat{B}^{ab}\hat{B}_{ba} - R_{ab}k^a k^b, \end{aligned} \tag{3.29}$$

where we have used (3.17), (3.22), and the symmetry property of the Riemann tensor. Additionally, we have

$$\begin{aligned} \hat{B}^{ab}\hat{B}_{ba} &= \left(\frac{1}{D-2}\theta h^{ab} + \hat{\sigma}^{ab} + \hat{\omega}^{ab}\right)\left(\frac{1}{D-2}\theta h_{ab} + \hat{\sigma}_{ab} - \hat{\omega}_{ab}\right) \\ &= \frac{1}{D-2}\theta^2 + \hat{\sigma}_{ab}\hat{\sigma}^{ab} - \hat{\omega}_{ab}\hat{\omega}^{ab}, \end{aligned} \tag{3.30}$$

As a result, we obtain a key equation called *Raychaudhuri's equation*:

$$\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} = -\frac{1}{D-2}\theta^2 - \hat{\sigma}_{ab}\hat{\sigma}^{ab} + \hat{\omega}_{ab}\hat{\omega}^{ab} - R_{ab}k^a k^b, \tag{3.31}$$

which is the master equation for many classical singularity theorems. Note that $\hat{\sigma}_{ab}\hat{\sigma}^{ab}$ and $\hat{\omega}_{ab}\hat{\omega}^{ab}$ are independent of l^a. By the same calculations as above, we obtain

$$\hat{B}^{ab}\hat{B}_{ab} = \frac{1}{D-2}\theta^2 + \hat{\sigma}_{ab}\hat{\sigma}^{ab} + \hat{\omega}_{ab}\hat{\omega}^{ab}. \tag{3.32}$$

Additionally, using (3.17), we have $\hat{B}^{ab}\hat{B}_{ba} = B^{ab}B_{ba}$ and $\hat{B}^{ab}\hat{B}_{ab} = B^{ab}B_{ab}$. As a result, we obtain

$$\hat{\sigma}_{ab}\hat{\sigma}^{ab} = B^{ab}B_{(ab)} - \frac{1}{D-2}\theta^2, \tag{3.33}$$

$$\hat{\omega}_{ab}\hat{\omega}^{ab} = B^{ab}B_{[ab]}. \tag{3.34}$$

This shows that the definition and the evolution of θ depend solely on the geodesics congruences and not the choice of Σ and l^a.

3.2 Null Hypersurfaces

A hypersurface $\mathcal{N}$ is a submanifold of codimension 1 in the spacetime manifold M. One way to define a hypersurface is to define a smooth scalar field $f : M \to \mathbb{R}$ such that $\mathcal{N}$ is the set of points $p \in M$ satisfying $f(p) = 0$. The vector field

$$\zeta^a = g^{ab}\nabla_b f \tag{3.35}$$

defined on the hypersurface $\mathcal{N}$ is a normal vector field. Indeed, for each point $p \in \mathcal{N}$ and any tangent vector $v^a \in T_p\mathcal{N} \subset T_pM$, we have

$$v^a\zeta_a = v^a\nabla_a f = v(f) = 0, \tag{3.36}$$

where we have used the fact that f is a constant function on $\mathcal{N}$. In this section, we consider a null hypersurface $\mathcal{N}$ where the normal vector field ζ^a is a null vector field on $\mathcal{N}$, i.e. $\zeta^a\zeta_a = 0$ everywhere on $\mathcal{N}$. For such a hypersurface, the normal vector field ζ^a is also tangent to $\mathcal{N}$. Surprisingly, the integral curves of ζ^a are null geodesics. In fact, if we extend the definition (3.35) of ζ^a to a neighborhood of $\mathcal{N}$, we have

$$\zeta^b\nabla_b\zeta_a = \zeta^b\nabla_b\nabla_a f = \zeta^b\nabla_a\nabla_b f = \zeta^b\nabla_a\zeta_b = \frac{1}{2}\nabla_a(\zeta^b\zeta_b). \tag{3.37}$$

Since the scalar function $\zeta^b\zeta_b$ vanishes everywhere on $\mathcal{N}$, its gradient $\nabla^a(\zeta^b\zeta_b)$ must be proportional to ζ^a on $\mathcal{N}$. Hence, there exists a function α on $\mathcal{N}$ such that $\nabla^a(\zeta^b\zeta_b) = 2\alpha\zeta^a$ and thus, ζ^a satisfies the geodesic equation

$$\zeta^b\nabla_b\zeta^a = \alpha\zeta^a. \tag{3.38}$$

This shows that the hypersurface $\mathcal{N}$ is a subset of a null geodesic congruence where each null geodesic on $\mathcal{N}$ is called a *generator* of $\mathcal{N}$. Furthermore, we can rescale the vector field ζ^a by a scalar function β so that the vector field $k^a = \beta\zeta^a$ satisfies the equation (3.1), i.e. each generator is affinely parametrized by λ. To form the null hypersurface $\mathcal{N}$, the vector field k^a on $\mathcal{N}$ must satisfy the Frobenius theorem, which leads to the following characterization of generators of $\mathcal{N}$.

Proposition 3.1. *If the congruence of null geodesics contains the generators of a null hypersurface $\mathcal{N}$, then $\hat{\omega}_{ab} = 0$ on $\mathcal{N}$. Conversely, if $\hat{\omega}_{ab} = 0$ everywhere, then the vector field k^a is everywhere hypersurface orthogonal, i.e. orthogonal to a family of null hypersurfaces.*

Proof. From (3.5) and (3.6), we have $B_{ab}k^b = B_{ba}k^b = 0$, which leads to

$$\hat{B}_{ab} = h_a{}^c h_b{}^d B_{cd} = B_{ab} + B_{ad}l^d k_b + B_{cb}l^c k_a + B_{cd}l^c l^d k_a k_b. \tag{3.39}$$

Then, the rotation can be expressed as

$$\hat{\omega}_{ab} = \hat{B}_{[ab]} = B_{[ab]} + B_{[a|d}k_{|b]}l^d + k_{[a|}B_{c|b]}l^c. \tag{3.40}$$

Hence, we obtain

$$k_{[a}\hat{\omega}_{bc]} = k_{[a}B_{bc]} + k_{[a}B_{b|d}k_{|c]}l^d + k_{[a}k_{b|}B_{d|c]}l^d = k_{[a}B_{bc]} = k_{[a}\nabla_b k_{c]}. \tag{3.41}$$

The vector field k^a is hypersurface orthogonal if and only if it satisfies $k_{[a}\nabla_b k_{c]} = 0$ according to the Frobenius theorem. Consequently, this condition is equivalent to

$$k_{[a}\hat{\omega}_{bc]} = 0. \tag{3.42}$$

It is clear that this condition is trivially satisfied if $\hat{\omega}_{ab} = 0$. Conversely, if this condition holds, then we take a contraction with l^a

$$0 = l^a k_{[a}\hat{\omega}_{bc]} = \frac{1}{3}l^a\left(k_a\hat{\omega}_{bc} + k_b\hat{\omega}_{ca} + k_c\hat{\omega}_{ab}\right) = -\frac{1}{3}\hat{\omega}_{bc} \tag{3.43}$$

where we use $k_a l^a = -1$ (3.9) and $\hat{B}_{ab}l^b = \hat{B}_{ba}l^b = 0$ (3.22). □

The expansion θ is similar to divergence of a vector field in multivariate calculus. When θ is positive at a point p, the geodesics close to p have a tendency to converge. Otherwise, if θ is negative at p, then the geodesics close to p tend to diverge. We will see more about this in section about conjugated points.

Heuristically, the cross-section of these geodesics expands when $\theta > 0$ and shrinks when $\theta < 0$. To make this precise, we introduce *Gaussian null coordinates* near $\mathcal{N}$. Consider again a spacelike hypersurface Σ intersecting each generator once on $\mathcal{N}$. The vector field k^a is normal to $\mathcal{S} = \mathcal{N} \cap \Sigma$, which shows that $\mathcal{S}$ is a spacelike surface of codimension 2. Therefore, we can find coordinates y^i $(i = 1, \ldots, D-2)$ on $\mathcal{S}$. We shift the parameter λ so that λ is constant on $\mathcal{S}$. Every point p on $\mathcal{N}$ can be assigned coordinates (v, y^i) so that there is a geodesic γ passing through p at $\lambda = v$ and intersecting $\mathcal{S}$ at a point with coordinates y^i. Next, we pick a null vector field l^a on $\mathcal{N}$ satisfying (3.9). In particular, we define l^a as a linear combination of a timelike normal vector of Σ and a spacelike normal vector of $\mathcal{S}$, which makes l^a normal to $\mathcal{S}$. Then, by using l^a as a initial condition on $\mathcal{N}$, we consider null geodesics μ emanating from $\mathcal{N}$ and affinely parametrized by λ' so that $\lambda' = 0$ on $\mathcal{N}$. Unless these geodesics intersect with each other, we can assign coordinates (r, v, y^i) to every spacetime point p near or on $\mathcal{N}$ so that there is a geodesic μ passing through p at $\lambda' = r$ and intersecting $\mathcal{N}$ at a point with coordinates (v, y^i). The coordinate basis vectors are given by

$$l^a = \left(\frac{\partial}{\partial r}\right)^a, \quad k^a = \left(\frac{\partial}{\partial v}\right)^a, \quad Y_i^a = \left(\frac{\partial}{\partial y^i}\right)^a. \tag{3.44}$$

Furthermore, since $l^a l_a = 0$ everywhere, we can show that

$$l^a \nabla_a (l_b k^b) = l_b l^a \nabla_a k^b = l_b k^a \nabla_a l^b = \frac{1}{2} = k^a \nabla_a (l^b l_b) = 0, \tag{3.45}$$

and similarly, $l^a \nabla_a (l_b Y_i^b) = 0$ for all $i = 1, \ldots, D-2$. As a result, since $l_a k^a = -1$ and $l_a Y_i^a = 0$ on $\mathcal{N}$, we have

$$l_a k^a = -1, \qquad l_a Y_i^a = 0 \tag{3.46}$$

everywhere. The expression of the metric in this coordinate system is

$$\mathrm{d}s^2 = g_{vv} \mathrm{d}v^2 - 2\mathrm{d}v\mathrm{d}r + g_{vi} \mathrm{d}v\mathrm{d}y^i + h_{ij} \mathrm{d}y^i \mathrm{d}y^j, \tag{3.47}$$

where $g_{vv} = k^a k_a$ and $g_{vi} = k_a Y_i^a$. Since $g_{vv} = g_{vi} = 0$ on $\mathcal{N}$, the metric restricted to $\mathcal{N}$ is

$$\mathrm{d}s^2|_{\mathcal{N}} = -2\mathrm{d}v\mathrm{d}r + h_{ij} \mathrm{d}y^i \mathrm{d}y^j. \tag{3.48}$$

The determinant of the metric on $\mathcal{N}$ is

$$g|_{\mathcal{N}} = \det(g_{\mu\nu}|_{\mathcal{N}}) = \begin{vmatrix} 0 & -1 & \\ -1 & 0 & \\ & & \boxed{h_{ij}} \end{vmatrix} = -\det(h_{ij}) = -h. \tag{3.49}$$

Then, the expansion θ on $\mathcal{N}$ is given by

$$\begin{aligned} \theta = \nabla_a k^a|_{\mathcal{N}} = \frac{1}{\sqrt{-g}} \frac{\partial}{\partial x^\mu} \left(\sqrt{-g} k^\mu \right) \Big|_{\mathcal{N}} &= \frac{1}{\sqrt{-g}} \frac{\partial}{\partial v} \left(\sqrt{-g} \right) \Big|_{\mathcal{N}} \\ &= \frac{1}{\sqrt{-g|_{\mathcal{N}}}} \frac{\partial}{\partial \lambda} \left(\sqrt{-g|_{\mathcal{N}}} \right) = \frac{1}{\sqrt{h}} \frac{\partial \sqrt{h}}{\partial \lambda}. \end{aligned} \tag{3.50}$$

where we replace v by λ in the last two steps because $v = \lambda$ is the affine parameter of generators on $\mathcal{N}$. In fact, since h_{ij} is the pullback of the metric $g_{\mu\nu}$ onto $\mathcal{S}$, the quantity $\sqrt{h}$ is the 'area' element on the surface $\mathcal{S}$ of constant λ. Hence, $\theta \, \mathrm{d}\lambda$ measures the fractional change of an infinitesimal area around a point on the cross-section $\mathcal{S}$. Indeed, by using the infinitesimal area $\mathcal{A} = \sqrt{h} \, \mathrm{d}^{D-2}y$, we can re-express the expansion θ as

$$\theta = \lim_{\mathcal{A} \to 0} \frac{1}{\mathcal{A}} \frac{\mathrm{d}\mathcal{A}}{\mathrm{d}\lambda}. \tag{3.51}$$

In the context of null hypersurfaces, the symmetric tensors h_{ab} and $\hat{B}_{ab}$ introduced in the last section are called the null *first* and *second fundamental forms*, respectively. In particular, $\hat{B}_{ab}$ is also called the *extrinsic curvature*. Furthermore, the subspace

$T_\perp$ at each point on the cross-section $\mathcal{S}$ is actually the tangent space of $\mathcal{S}$ at that point. Therefore, the first fundamental form h_{ab}, which becomes h_{ij} when being pullbacked to $\mathcal{S}$, acts as the induced metric on $\mathcal{S}$. Moreover, the extrinsic curvature $\hat{B}_{ab}$ measures how the null hypersurface $\mathcal{N}$ is curved within the ambient spacetime M. This can be seen from the fact that $\hat{B}_{ab}$ determines how the spatial separation $\hat{\eta}^a$ between generators on hypersurfaces changes.

3.3 Null Convergence Condition and Energy Conditions

Consider a null hypersurface $\mathcal{N}$. Proposition 3.1 shows that the rotation tensor ω_{ab} must vanish on $\mathcal{N}$. Then, the Raychaudhuri equation (3.31) becomes

$$\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} = -\frac{1}{D-2}\theta^2 - \hat{\sigma}_{ab}\hat{\sigma}^{ab} - R_{ab}k^a k^b. \tag{3.52}$$

Note that $\hat{\sigma}_{ab}\hat{\sigma}^{ab}$ is always non-negative because it is the sum of squares of ${\sigma^a}_b$'s eigenvalues. If we assume the *null convergence condition*, which states that $R_{ab}K^aK^b \geq 0$ for all null vector K^a, then the equation (3.52) gives

$$\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} \leq 0, \tag{3.53}$$

which is called the *classical focusing theorem* [17]. This shows that the generators on $\mathcal{N}$ are either parallel to each other or converging if the null convergence condition holds. However, the Raychaudhuri equation provides much more information than that. Indeed, the following lemma shows that once the generators start to converge at some point, then the convergence inevitably happens within a finite affine distance from that point.

Lemma 3.2. *Let k^a be the tangent of generators of a null hypersurface. Assume the null convergence condition everywhere on this hypersurface so that $R_{ab}k^ak^b \geq 0$. If the expansion θ takes a negative value θ_0 at a point $\gamma(0)$ on the generator γ, then θ goes to $-\infty$ along that generator at an affine distance $\lambda_\infty \leq (D-2)/|\theta_0|$.*

Proof. Given the null convergence condition, the Raychaudhuri equation implies that

$$\begin{aligned} \frac{\mathrm{d}\theta}{\mathrm{d}\lambda} \leq -\frac{1}{D-2}\theta^2 \quad &\Leftrightarrow \quad \int_0^{\lambda_\infty} \mathrm{d}\lambda \leq -(D-2)\int_{\theta_0}^{-\infty} \frac{\mathrm{d}\theta}{\theta^2} \\ &\Leftrightarrow \quad \lambda_\infty \leq -\frac{D-2}{\theta_0}, \end{aligned} \tag{3.54}$$

which gives the upper bound $(D-2)/|\theta_0|$ for λ_∞. □

This result plays an important role in the proof of Penrose singularity theorem.

Next, we want to discuss physical situations that lead to the null convergence condition. For now, assume that spacetime is 4-dimensional, i.e. $D = 4$. Using Einstein's equation with a cosmological constant

$$R_{ab} - \frac{1}{2}Rg_{ab} + \Lambda g_{ab} = 8\pi G T_{ab}, \tag{3.55}$$

we obtain

$$\left(R_{ab} - \frac{1}{2}Rg_{ab} + \Lambda g_{ab}\right) g^{ab} = 8\pi G T_{ab} g^{ab} \quad \Longleftrightarrow \quad R = -(8\pi G T - 4\Lambda). \tag{3.56}$$

Then, using Einstein's equation again, we obtain

$$\begin{aligned} R_{ab} = 8\pi G T_{ab} + \frac{1}{2}Rg_{ab} - \Lambda g_{ab} &= 8\pi G T_{ab} - \frac{1}{2}(8\pi G T - 4\Lambda) g_{ab} - \Lambda g_{ab} \\ &= 8\pi G \left(T_{ab} - \frac{1}{2}T g_{ab} + \frac{\Lambda}{8\pi G} g_{ab}\right). \end{aligned} \tag{3.57}$$

Finally, we for every null vector k^a,

$$R_{ab}k^a k^b = 8\pi G \left(T_{ab} - \frac{1}{2}T g_{ab} + \frac{\Lambda}{8\pi G} g_{ab}\right) k^a k^b = 8\pi G T_{ab} k^a k^b. \tag{3.58}$$

The null convergence condition holds if we impose the following *energy conditions* on the stress-energy tensor:

(i) *Null Energy Condition (NEC)*: $T_{ab}k^a k^b \geq 0$ for all null vectors k^a;

(ii) *Weak Energy Condition (WEC)*: $T_{ab}\xi^a \xi^b \geq 0$ for all timelike vectors ξ^a;

(iii) *Strong Energy Condition (SEC)*: $\left(T_{ab} - \frac{1}{2}T g_{ab}\right)\xi^a \xi^b \geq 0$ for all timelike vectors ξ^a.

The connection between the null convergence condition and the NEC can be seen directly from (3.58). The physical meaning of the NEC is that radiation energy densities are always positive. The WEC, which imposes positivity on energy densities, implies the NEC by the continuity of the function $v^a \mapsto T_{ab}v^a v^b$. Similarly, the SEC imposing attractiveness on gravity also implies the NEC by the continuity of $v^a \mapsto \left(T_{ab} - \frac{1}{2}T g_{ab}\right) v^a v^b$, but it is only appropriate in a universe with a vanishing cosmological constant. However, SEC is needed in many important theorems of general relativity such as Penrose-Hawking singularity theorem (1970) [8].

3.4 Conjugate Points

Consider again a general congruence of null geodesics. Using (3.20) and (3.28), the acceleration of $\hat{\eta}_a$ is given by

$$
\begin{aligned}
k^c\nabla_c(k^b\nabla_b\hat{\eta}_a) = k^c\nabla_c(\hat{B}_{ab}\hat{\eta}^b) &= \hat{B}_{ab}k^c\nabla_c\hat{\eta}^b + \hat{\eta}^b k^c\nabla_c\hat{B}_{ab} \\
&= \hat{B}_{ab}\hat{B}^b{}_c\hat{\eta}^c - \hat{\eta}^b\hat{B}_{ac}\hat{B}^c{}_b - \hat{\eta}^b h_a{}^e h_b{}^d R_{dcef}k^c k^f \\
&= -h_a{}^e R_{dcef}k^c k^f\hat{\eta}^d
\end{aligned}
\tag{3.59}
$$

Using the fact that h_{ab} is parallel transported along the geodesics and $h^a{}_b\hat{\eta}^b = \hat{\eta}^a$, we obtain

$$
k^c\nabla_c(k^b\nabla_b\hat{\eta}_a) = -R_{cbda}\hat{\eta}^b k^c k^d. \tag{3.60}
$$

The *geodesic deviation equation* (3.60) has an advantage over (3.20). That is, the RHS of (3.60) depends only on tangent k^a of *one* null geodesic, whereas the tensor $\hat{B}_{ab} = \nabla_b k_a$ in (3.20) depends on how a *group* of geodesics moves.

Now, consider a single null geodesic γ with a tangent k^a. A vector field $\hat{\eta}^a$ along γ is called a *Jacobi field* if it satisfies the geodesic deviation equation (3.60). The vector $\hat{\eta}^a$ at each point of γ belongs to the $(D-2)$-dimensional subspace $T_\perp$. We choose an orthonormal basis $\{E_i^a\}_{i=1}^{D-2}$ of $T_\perp$ at one point of γ and extend it over γ by parallel transporting each E_i^a along γ. Then, any Jacobi field $\hat{\eta}^a$ along γ can be expressed as

$$
\hat{\eta}^a = \sum_{i=1}^{D-2}\hat{\eta}^i E_i^a. \tag{3.61}
$$

Substituting this expression into (3.60), we obtain a system of linear differential equations for the coefficients $\hat{\eta}^i$ $(i = 1, \dots, D-2)$:

$$
\frac{\mathrm{d}^2\hat{\eta}^i}{\mathrm{d}\lambda^2} = -\sum_{j=1}^{D-2} R_{0j0i}\hat{\eta}^j, \tag{3.62}
$$

where $R_{0j0i} = R_{cbda}E_j^b k^c k^d E_i^a$. Given initial values $\hat{\eta}^i(0)$ and $\mathrm{d}\hat{\eta}^i/\mathrm{d}\lambda|_{\lambda=0}$, the solution of (3.62) is unique.

For the rest of this section, consider a smooth spacelike surface $\mathcal{S}$ of codimension 2 and a family of null geodesics orthogonal to $\mathcal{S}$ at every point on $\mathcal{S}$. These geodesics form a null hypersurface $\mathcal{N}$ emanating from $\mathcal{S}$. Let γ be one of these geodesics starting from a point $p = \gamma(0) \in \mathcal{S}$. The initial velocity $\mathrm{d}\hat{\eta}^i/\mathrm{d}\lambda|_{\lambda=0}$ is given by (3.20), i.e.

$$
\left.\frac{\mathrm{d}\hat{\eta}^i}{\mathrm{d}\lambda}\right|_{\lambda=0} = \sum_{j=1}^{D-2}\hat{B}_{ij}\hat{\eta}^j(0), \tag{3.63}
$$

where $\hat{B}_{ij} = \hat{B}_{ab} E^a_i E^b_j$. Then, the solution of the linear equations (3.62) with this initial condition can be expressed as a linear combination of only $\hat{\eta}^i(0)$:

$$\hat{\eta}^i(\lambda) = \sum_{j=1}^{D-2} A_{ij}(\lambda)\hat{\eta}^j(0), \tag{3.64}$$

where A_{ij} is a $(D-2)\times(D-2)$ matrix satisfying the equation

$$\frac{\mathrm{d}A_{ij}}{\mathrm{d}\lambda} = \sum_{k=1}^{D-2} \hat{B}_{ik} A_{kj}, \tag{3.65}$$

$$\frac{\mathrm{d}^2 A_{ij}}{\mathrm{d}\lambda^2} = -\sum_{k=1}^{D-2} R_{0k0i} A_{kj}, \tag{3.66}$$

and $A_{ij}(0) = I_{D-2}$ where I_{D-2} is a $(D-2)\times(D-2)$ identity matrix. Using Jacobi's formula and (3.65), we have

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\det \mathbf{A} = \det \mathbf{A}\ \ \mathrm{Tr}\left(\mathbf{A}^{-1}\frac{\mathrm{d}\mathbf{A}}{\mathrm{d}\lambda}\right) = \det \mathbf{A}\ \ \mathrm{Tr}\left(\mathbf{A}^{-1}\hat{\mathbf{B}}\mathbf{A}\right) = \det \mathbf{A}\ \ \mathrm{Tr}\Big(\hat{\mathbf{B}}\Big). \tag{3.67}$$

Moreover, we have

$$\mathrm{Tr}\Big(\hat{\mathbf{B}}\Big) = \sum_{i=1}^{D-2} \hat{B}_{ii} = \hat{B}_{ab}\sum_{i=1}^{D-2} E^a_i E^b_i = \hat{B}_{ab}h^{ab} = \theta. \tag{3.68}$$

As a result, we obtain the relation between the matrix A_{ij} and the expansion θ:

$$\theta = \frac{1}{\det \mathbf{A}}\frac{\mathrm{d}}{\mathrm{d}\lambda}(\det \mathbf{A}). \tag{3.69}$$

A point q *conjugated* to $\mathcal{S}$ along γ is defined as a point on γ such that there exists a Jacobi field $\hat{\eta}^a$ satisfying the initial condition (3.63) on $\mathcal{S}$ and vanishing at q but not identically zero along γ. There is a point $q = \gamma(\lambda_\infty)$ conjugated to $\mathcal{S}$ along γ if and only if the matrix $A_{ij}(\lambda_\infty)$ is not invertible, i.e. mapping a non-zero vector $\hat{\eta}^i(0)$ to the zero vector. In addition, the equation (3.66) shows that $\mathrm{d}/\mathrm{d}\lambda\,(\det \mathbf{A})$ is always finite when the curvature R_{k00i} is finite. Therefore, using (3.69), we conclude that the expansion θ goes to $-\infty$ at the conjugate point q. Combining this information with Lemma 3.2, we obtain the following proposition.

Proposition 3.3. *Let (M, g_{ab}) be a spacetime satisfying $R_{ab}k^a k^b \geq 0$ for all null vectors k^a. Let $\mathcal{S}$ be a smooth spacelike submanifold of codimension 2 such that the expansion θ of null geodesics orthogonal to $\mathcal{S}$ has a negative value θ_0 at a point $p \in \mathcal{S}$. Then, there exists a point q conjugate to S along the null geodesic γ passing through $p = \gamma(0)$ within the range $0 < \lambda \leq (D-2)/|\theta_0|$ of the affine parameter.*

Hawking and Ellis (1973) [6] shows that if there is a conjugate point on a null geodesic, it is possible to make a small variation of the curve so that it becomes a timelike curve.

Proposition 3.4. *If there is a point q conjugate to $\mathcal{S}$ along γ between $\mathcal{S}$ and p, then there is a variation of γ which gives a timelike curve from $\mathcal{S}$ to p.*

Proof. See Proposition 4.5.14 in [6] for more details. □

Corollary 3.5. *Let (M, g_{ab}) be a globally hyperbolic spacetime and let S be a compact, orientable, two-dimensional spacelike submanifold of M. Then, every event $p \in \partial I^+(S)$ lies on a future directed null geodesic starting from S which is orthogonal to S and has no point conjugate to S between S and p.*

Proof. When there is a conjugate point q on a null geodesic between $\mathcal{S}$ and p, then for any point r on the geodesic between q and p, it is possible to deform the geodesic connecting $\mathcal{S}$ and r into a timelike curve. Therefore, these points r belong to $I^+(\mathcal{S})$ □

3.5 Penrose Singularity Theorem

In the last section, we established some results about null geodesics orthogonal to a smooth spacelike surface of codimension 2. Let us describe this kind of surfaces in more details. A spacelike surface $\mathcal{T}$ of codimension 2 is a set of points p satisfying two equations $f_1(p) = 0$ and $f_2(p) = 0$, where f_1 and f_2 are smooth functions on M such that their gradients $\nabla_a f_1$ and $\nabla_a f_2$ are non-vanishing and linearly independent at every point on $\mathcal{T}$ and

$$g^{ab}(\nabla_a f_1 + \mu \nabla_a f_2)(\nabla_b f_1 + \mu \nabla_b f_2) = 0 \tag{3.70}$$

for two distinct values of μ. Then, all tangent vectors on $\mathcal{T}$ are spacelike because they are orthogonal to two linearly independent null vectors. Let k^a and l^a be two future-directed null vector fields normal to $\mathcal{T}$ and proportional to $(\nabla^a f_1 + \mu \nabla^a f_2)$ for the two distinct values of μ so that $k^a l_a = -1$. Moreover, suppose that k^a and l^a are tangent to congruences of "outgoing" and "ingoing" null geodesics, respectively. A compact surface $\mathcal{T}$ of codimension 2 is called a *trapped surface* if both expansions $\theta^{(\mathrm{out})}$ and $\theta^{(\mathrm{in})}$ of these null geodesics are negative everywhere on $\mathcal{T}$. The Penrose singularity theorem [5] states that singularity is inescapable once a trapped surface forms in a spacetime satisfying some reasonable conditions. Compared Penrose's original proof, we present a more precise proof of Penrose's theorem as given in [1, 6].

Theorem 3.6 (Penrose, 1965)**.** *A spacetime (M, g_{ab}) is null geodesically incomplete if:*

(1) $R_{ab}k^a k^b \geq 0$ for every null vector k^a;

(2) There is a non-compact Cauchy surface $\mathcal{C}$*;*

(3) There is a trapped surface $\mathcal{T}$*.*

Remark 3.7. A spacetime is *null geodesically incomplete* if there exists a null geodesic that cannot be extended to arbitrary values of its affine parameter, i.e. the affine parameter approaches a finite value when it either increases or decreases.

Proof. Assume by contradiction that spacetime is null geodesically complete. By conditions (1), (3), and Proposition 3.3, every future-directed ingoing/outgoing null geodesic orthogonal to the trapped surface $\mathcal{T}$ runs into a conjugate point within an affine distance $(D-2)/|\theta_0^{(\text{in})/(\text{out})}(p)|$ starting from every point $p \in \mathcal{T}$ where $\theta_0^{(\text{in})/(\text{out})}(p)$ is the expansion of the ingoing/outgoing congruence at p. These null geodesics emanated from $\mathcal{T}$ form the boundary $\partial I^+(\mathcal{T})$ of the chronological future until when they terminate on $\partial I^+(\mathcal{T})$ before or at the conjugate points (Corollary 3.5). However, these geodesics can be extended from their future endpoints on $\partial I^+(\mathcal{T})$. In addition, there exists a minimum value $\theta_{\text{min}}^{(\text{in})/(\text{out})}$ of $\theta_0^{(\text{in})/(\text{out})}$ because $\mathcal{T}$ is compact. Define $\lambda_{\text{max}}^{(\text{in})/(\text{out})} = (D-2)/|\theta_{\text{min}}^{(\text{in})/(\text{out})}|$. Then, let us define a continuous map $f^{(\text{in})} : \mathcal{T} \times [0, \lambda_{\text{max}}^{(\text{in})}] \to M$ which sends a point p on $\mathcal{T}$ to a point $\gamma(\lambda)$ $\left(0 \leq \lambda \leq \lambda_{\text{max}}^{(\text{in})}\right)$ on the ingoing null geodesic γ starting from p. The map $f^{(\text{out})} : \mathcal{T} \times [0, \lambda_{\text{max}}^{(\text{out})}] \to M$. Since $\mathcal{T}$ and closed intervals are compact, the set

$$K = f^{(\text{in})}\left(\mathcal{T} \times [0, \lambda_{\text{max}}^{(\text{in})}]\right) \bigcup f^{(\text{out})}\left(\mathcal{T} \times [0, \lambda_{\text{max}}^{(\text{out})}]\right) \tag{3.71}$$

is also compact. As stated above, the boundary $\partial I^+(\mathcal{T})$ of the future of $\mathcal{T}$ is a subset of K. Since $\partial I^+(\mathcal{T})$ is closed due to Proposition 2.1, it is compact as a closed subset of a compact set.

As a final step, we show the fact that $\partial I^+(\mathcal{T})$ is compact leads to a contradiction with condition (2). First, there exists a continuous timelike vector field t^a on M because M is time-orientable. This vector field generates a family of continuous and inextendible timelike curves each of which intersects the Cauchy surface $\mathcal{C}$ exactly once. Furthermore, since $\partial I^+(\mathcal{T})$ is achronal due to Proposition 2.1, each of these timelike curves intersects $\partial I^+(\mathcal{T})$ at most once. Therefore, we can define a continuous map $\pi : \partial I^+(\mathcal{T}) \to \mathcal{C}$ which sends a point $p \in \partial I^+(\mathcal{T})$ to the intersection of $\mathcal{C}$ and the timelike curve generated by t^a passing through p. Then, the image $\pi(\partial I^+(\mathcal{T}))$ on $\mathcal{C}$ is compact and therefore, it is closed. Moreover, if we restrict the codomain of π to the image $\pi(\partial I^+(\mathcal{T}))$, then π becomes a homeomorphism between $\partial I^+(\mathcal{T})$ and $\pi(\partial I^+(\mathcal{T}))$. Since $\partial I^+(\mathcal{T})$ is also a submanifold of M due to Proposition 2.1, the set $\pi(\partial I^+(\mathcal{T}))$ must be an open subset of $\mathcal{C}$. Unless $\mathcal{C}$ is disconnected which is unphysical, we must have $\mathcal{C} = \pi(\partial I^+(\mathcal{T}))$ because $\mathcal{C}$ is the only non-empty subset of itself that is both open and closed. However, this implies that $\mathcal{C}$ is compact which contradicts the condition (2). Therefore, our assumption that spacetime is null geodesically complete must be false. □

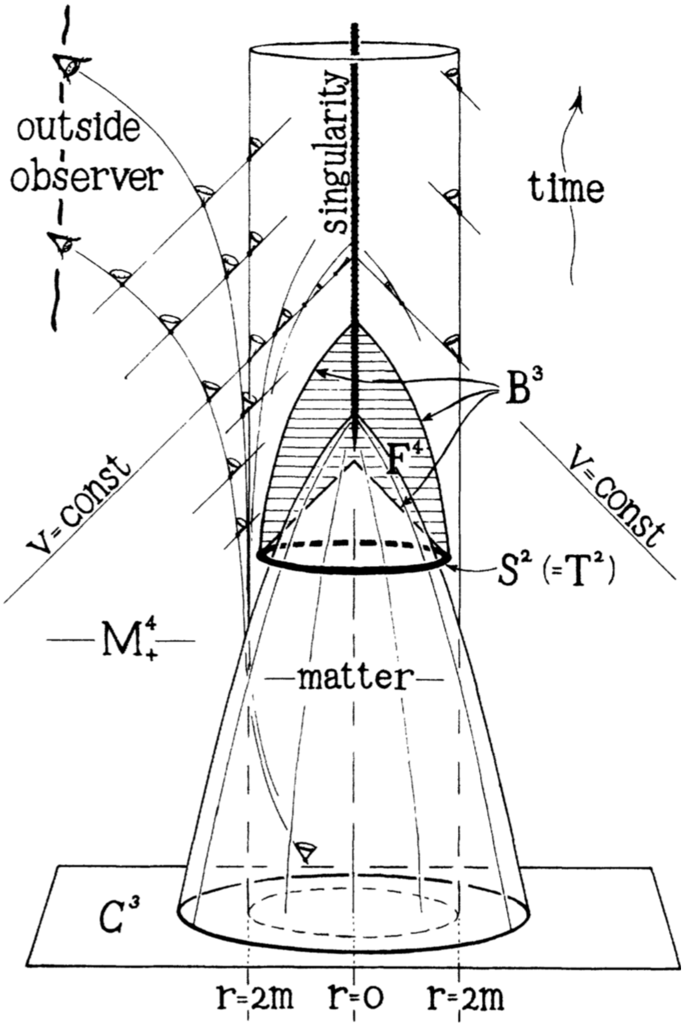

FIGURE 3.1: In a spherically symmetric gravitational collapse, a trapped surface T^2 is a 2-sphere S^2 from which the ingoing and the outgoing light form a compact set B^3 (Source: Penrose (1965) [5]).

Chapter 4

Penrose-Wall Singularity Theorem

4.1 The Second Law of Thermodynamics

4.1.1 The ordinary second law

The Ordinary Second Law (OSL) of Thermodynamics states that the entropy of a closed non-gravitating system must be non-decreasing over time. In this discussion, the entropy of a quantum system in a state represented by a density operator ρ is given by the von Neumann entropy

$$S = -\operatorname{tr}(\rho \ln \rho). \tag{4.1}$$

Then, the OSL can be understood in two ways depending on the accessible information about the dynamics of the system. In the fine-grained picture, an observer has unlimited access to all information about the system and therefore knows exactly which unitary transformation $U(t)$ determines the evolution of the system

$$\rho(t) = U(t)\rho(0)U^\dagger(t). \tag{4.2}$$

In this picture, the entropy is a constant over time due to the cyclic property of the trace, i.e.

$$\begin{aligned} S(\rho(t)) = -\operatorname{tr}\Big[U(t)\rho(0)\ln\rho(0)U^\dagger(t)\Big] &= -\operatorname{tr}\Big[U^\dagger(t)U(t)\rho(0)\ln\rho(0)\Big] \\ &= -\operatorname{tr}[\rho(0)\ln\rho(0)] = S(\rho(0)). \end{aligned} \tag{4.3}$$

This shows that the OSL is trivial in the fine-grained picture. On the other hand, observers in the coarse-grained picture are ordinary human beings who can only observe the system in a coarse-grained way. Due to their ignorance, the best they can do is to build a probabilistic model of the dynamics. That is, the unitary operator U determining the state of the system at the time t obeys a probability distribution with a probability density $\mu(U)$ satisfying

$$\int_U \mu(U)\, \mathrm{d}U = 1. \tag{4.4}$$

Then, the observed state of the system at the time t starting from a state ρ at $t = 0$ is the averaged result of all possible dynamics:

$$T(\rho) = \int_U U\rho U^\dagger \mu(U)\, \mathrm{d}U\,. \tag{4.5}$$

Then, it is possible to prove that the entropy of the state $T(\rho)$ is no smaller than the entropy of ρ, i.e.

$$S(T(\rho)) \geq S(\rho). \tag{4.6}$$

For the proof, see the appendix in [16]. However, we will consider only the fine-grained entropy in the later sections.

4.1.2 The generalized second law

The Generalized Second Law (GSL) was first proposed by Bekenstein in 1972 [10] to generalize the OSL in the context of black hole thermodynamics. To compensate for the entropy of matter lost inside a black hole, he proposed that the total entropy must include the black hole entropy which is observable to an outside observer since the entropy is proportional to the surface area of the black hole. Generally, consider a compact surface $\mathcal{T}$ of codimension 2 on a partial Cauchy surface Σ. The surface $\mathcal{T}$ divides Σ into 2 regions where we designate the noncompact region as the exterior and other one as the interior (see Figure 4.1). The *generalized entropy* is then defined as

$$S_{\text{gen}} = \frac{\langle A \rangle}{4\hbar G} + S_{\text{out}} + Q, \tag{4.7}$$

$$S_{\text{out}} = -\operatorname{tr}(\rho_{\text{out}} \ln \rho_{\text{out}}), \tag{4.8}$$

where Q is the subleading correction term due to some unknown theory of quantum gravity.

The state ρ_{out} is the partial trace $\operatorname{tr}_{\text{in}}(\rho_{\text{total}})$ where ρ_{total} is the state of the composite system including matter in both regions. If ρ_{total} is a pure state $|\psi\rangle\langle\psi|$, then the entropies of the interior and the exterior are equal, i.e. $S_{\text{in}} = S_{\text{out}}$. In this case, this entropy measures the degree of entanglement between the inside and the outside.

The factor $\langle A \rangle$ in the first term of (4.7) is the expectation value of the area of $\mathcal{T}$. The randomness of the quantum matter outside $\mathcal{T}$ can cause the area of $\mathcal{T}$ to fluctuate. For example, there exists a spacetime where the formation of a black hole depends on the outcome of a quantum event [22]. In this scenario, the area of the horizon at a certain time can take a value of either 0 or a finite value. Therefore, the area of $\mathcal{T}$

in general should be understood as a quantum operator with the expectation value

$$\langle A \rangle = \mathrm{tr}(A\rho_{\mathrm{out}}). \tag{4.9}$$

The appearance of $\langle A \rangle$ in the first term of (4.7) is justified by the path integral formulation used in [22].

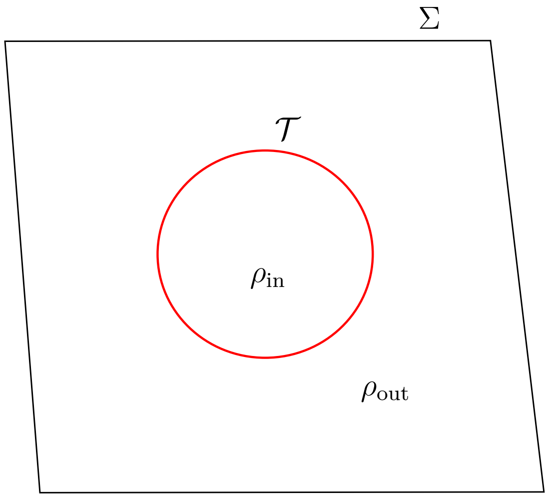

FIGURE 4.1: A compact surface $\mathcal{T}$ of codimension 2 separates a partial Cauchy surface Σ into 2 regions. We consider entropy of matter only in one region.

Beside black hole horizons, the GSL is also applicable to de Sitter horizons and Rindler horizons. Generally, its applicability extends to all *future causal horizons* each of which is defined as the boundary of the past of a timelike curve extending to a future infinity, which is also called a *future-infinite* timelike curve [14]. For example, a future event horizon $\mathcal{H}^+$ of a black hole in a flat universe is the boundary of past of a non-accelerating observer moving on a timelike curve γ extending to the timelike infinity i^+, i.e. $\mathcal{H}^+ = \partial I^-(\gamma)$. If this observer accelerates indefinitely, then $\partial I^-(\gamma)$ becomes a Rindler horizon. In a universe with a positive cosmological constant, every observer, accelerating or not, has a cosmological horizon, also called a de Sitter horizon. As formulated by Jacobson and Parentani in 2003 [14], the GSL can be stated as follows.

Conjecture 4.1 (GSL)**.** *Consider any future-infinite timelike worldline W_{fut} of an observer. The future causal horizon of this observer is defined as $H_{\mathrm{fut}} = \partial I^-(W_{\mathrm{fut}})$, the boundary of the past of the observer. Given any two partial Cauchy surfaces Σ and Σ' where the latter is nowhere to the past of the former (i.e. $\Sigma' \cap I^-(\Sigma) = \emptyset$),* *the GSL states that*

$$S_{\text{gen}}(\Sigma' \cap I^-(W_{\text{fut}})) \geq S_{\text{gen}}(\Sigma \cap I^-(W_{\text{fut}})). \tag{4.10}$$

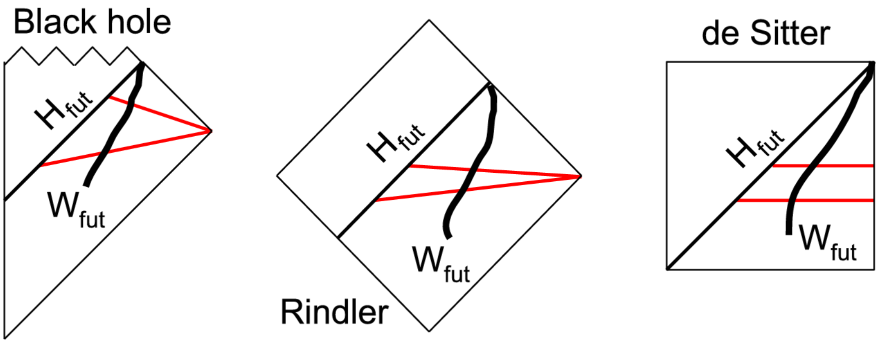

FIGURE 4.2: Conformal diagrams show 3 examples of future causal horizons [16]. Two red lines in each diagram show two partial Cauchy surfaces Σ and Σ'

Note that this conjecture is also applicable to horizons which are the boundaries of the pasts of null curves by continuity. Some proofs of Conjecture 4.1 in some regimes exist, although they all have limitations [23, 24]. Thus, GSL remains a conjecture, and it is an open question whether GSL holds generally in quantum gravity.

4.1.3 Quantum expansion

Similar to Hawking's area theorem [9] for event horizons of black holes, Penrose (1974) [25] and Gibbons & Hawking (1977) [13] argued that the following area theorem for general future causal horizons holds. Their arguments use some formulations of the Cosmic Censorship Conjecture so that no naked singularity occurs. The following statement based on [13] uses such a formulation.

Theorem 4.2 (Area Theorem for Future Causal Horizons)**.** *Consider an observer following a future-infinite timelike curve γ. Suppose that what this observer can see at late times can be predicted from a partial Cauchy surface Σ, i.e. $I^-(\gamma) \cap J^+(\Sigma)$ is contained in $D^+(\Sigma)$. Suppose further that the causal horizon $\mathcal{H} = \partial I^+(\gamma)$ is also contained in $D^+(\Sigma)$ so that it is predictable. Assume the NEC, i.e. $T_{ab}k^a k^b \geq 0$ for every null vector k^a. Then, the expansion θ on $\mathcal{H}$ is non-negative.*

Proof. Since the generators of $\mathcal{H}$ do not run into any naked singularity, they are all complete in the future direction. Suppose by contradiction that θ is negative at a point p on a surface $\mathcal{S}$ of codimension 2 inside $\mathcal{H}$. By Lemma 3.2, the generator through p runs into a conjugate point q within a finite affine length. Every point on this generator beyond q can be connected to $\mathcal{S}$ by a timelike curve (Proposition 3.3). This implies that these points must be outside $\mathcal{H}$, otherwise the achronality of $\mathcal{H}$ is violated. However, this also contradicts the fact that every generator of $\mathcal{H}$ has an infinite affine length into the future. $\square$

This statement motivates the replacement of the NEC by the GSL in the Penrose-Wall singularity theorem. Indeed, we have known that the Bekenstein-Hawking entropy (the first term in (4.7)) is proportional to the area of the horizon at one time. Hence, the formula (3.51) can be rewritten as

$$\theta = \lim_{\mathcal{A}\to 0} \frac{4G\hbar}{\mathcal{A}} \frac{\mathrm{d}S_{\mathrm{BH}}}{\mathrm{d}\lambda}. \tag{4.11}$$

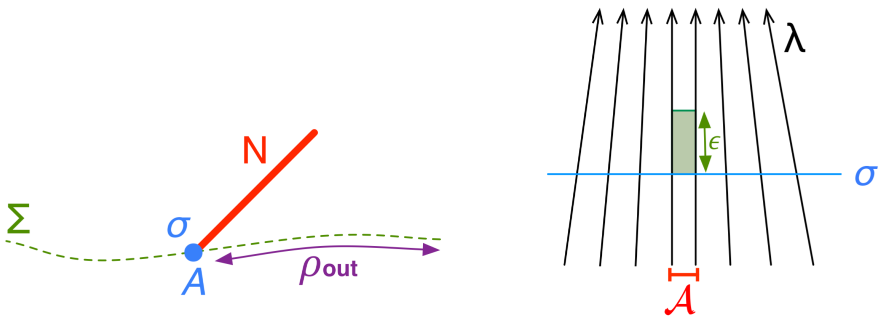

FIGURE 4.3: The diagrams show the setup in the definition of quantum expansions (Source: Bousso et al. (2016) [17]).

We can rewrite the GSL similarly by defining **quantum expansion**. Consider a null hypersurface $\mathcal{N}$ and a partial Cauchy surface intersecting $\mathcal{N}$ at a surface σ of codimension 2. Also, consider an infinitesimal area $\mathcal{A}$ around a point p on σ. Suppose that this area $\mathcal{A}$ is deformed by an infinitesimal affine length ϵ along generators of $\mathcal{N}$. This deformation of σ corresponds to a new partial Cauchy surface Σ' slicing $\mathcal{N}$, where the generalized entropy is changed by an amount δS_{gen}. Then, the quantum expansion is defined as

$$\Theta^{(\mathcal{N})}[\Sigma; p] = \lim_{\mathcal{A}\to 0} \lim_{\epsilon\to 0} \frac{4G\hbar}{\mathcal{A}} \frac{\delta S_{\mathrm{gen}}}{\epsilon} = \lim_{\mathcal{A}\to 0} \frac{4G\hbar}{\mathcal{A}} \frac{\mathrm{d}S_{\mathrm{gen}}}{\mathrm{d}\lambda} = \theta^{(\mathcal{N})}|_p + \lim_{\mathcal{A}\to 0} \frac{4G\hbar}{\mathcal{A}} \frac{\mathrm{d}S_{\mathrm{out}}}{\mathrm{d}\lambda}.$$

This notion of quantum expansions was introduced by Bousso et al. (2016) [17] in the context of the Quantum Focusing Conjecture (QFC), the quantum generalization of the Classical Focusing Theorem (3.53). Using this new notion, the GSL can be reformulated as follows.

Conjecture 4.3 (GSL). *Consider any future-infinite timelike worldline W_{fut} of an observer with the future causal horizon $H_{\mathrm{fut}} = \partial I^-(W_{\mathrm{fut}})$. Given any partial Cauchy surface Σ and any point $p \in H_{\mathrm{fut}} \cap \Sigma$, the GSL states that $\Theta^{(H_{\mathrm{fut}})}[\Sigma; p] \geq 0$.*

Unlike the classical expansion $\theta^{(\mathcal{N})}$ which depends only on $\sigma = \mathcal{N} \cap \Sigma$, the quantum expansion $\Theta^{(\mathcal{N})}$ also depends on the exterior of σ on Σ due to the entropy S_{out}. This quantum correction vanishes in the classical limit $\hbar \to 0$, which turns the GSL into the area theorem. In addition, this modern formulation of GSL avoids the problem of subtracting infinity from infinity when calculating the change $\Delta S_{\rm gen}$ in the total generalized entropy in the case that the area of the horizon $H_{\rm fut} \cap \Sigma$ such as a Rindler horizon is infinite. This way of writing the GSL can also be found in the reprint [18].

4.2 Generalized Thermodynamics Theorems

4.2.1 A monotonicity property of the generalized entropy

The following theorem, which is also called the touching lemma [18, 26], is the key result leading to the singularity theorem.

Theorem 4.4. *Let $\mathcal{N}$ and $\mathcal{M}$ be future null hypersurfaces of codimension 1 each of which divides spacetime into two regions, an 'interior' Int and an 'exterior' Ext. That is, the exterior of these surfaces contains the pasts of the surfaces. Let $\mathcal{M}$ be either within the interior $\mathrm{Int}(\mathcal{N})$ or on $\mathcal{N}$ everywhere, i.e. $\mathcal{M} \cap \mathrm{Ext}(\mathcal{N}) = \emptyset$.*

Let there be a null geodesic g which lies on both $\mathcal{N}$ and $\mathcal{M}$, and a partial Cauchy surface Σ which intersects g at g_0. Assume that in some neighborhood of g_0, the spacetime is semi-classical, and $\mathcal{N}$ and $\mathcal{M}$ are both smooth. Very close to g_0, the null hypersurfaces $\mathcal{N}$ and $\mathcal{M}$ will nearly coincide, but they may be separated by a small proper distance f.

For any neighborhood in the vicinity of g_0, there exists a way to evolve Σ forwards in time in that neighborhood to a new slice Σ', such that the generalized entropy increases faster on $\mathcal{M}$ than on $\mathcal{N}$:

$$\Delta S_{\rm gen}(\Sigma \cap \mathrm{Ext}(\mathcal{M})) \geq \Delta S_{\rm gen}(\Sigma \cap \mathrm{Ext}(\mathcal{N})), \tag{4.12}$$

where

$$\Delta S_{\rm gen}(\Sigma \cap \mathrm{Ext}(\cdot)) = S_{\rm gen}(\Sigma' \cap \mathrm{Ext}(\cdot)) - S_{\rm gen}(\Sigma \cap \mathrm{Ext}(\cdot)). \tag{4.13}$$

Or equivalently,

$$\Theta^{(\mathcal{M})}[\Sigma; g_0] \geq \Theta^{(\mathcal{N})}[\Sigma; g_0]. \tag{4.14}$$

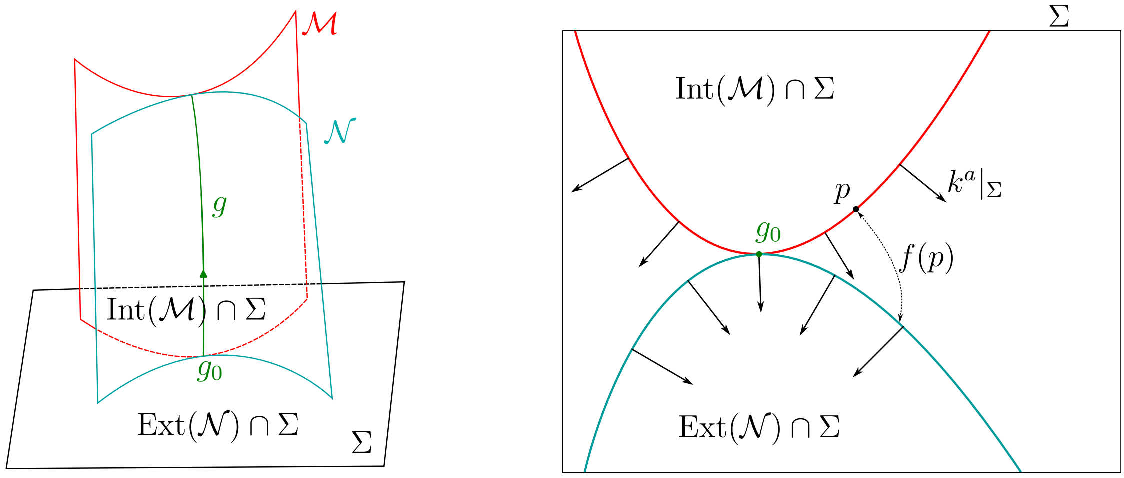

FIGURE 4.4: Two future null hypersurfaces $\mathcal{N}$ and $\mathcal{M}$ touch at the generator g.

As provided in [16, 27], the proof of the following lemma, which pertains to the Bekenstein-Hawking term of the generalized entropy, is given as follows.

Lemma 4.5. *In any small neighborhood of g_0, either there is a point p at which $\theta^{(\mathcal{M})} > \theta^{(\mathcal{N})}$ or else $\mathcal{N}$ and $\mathcal{M}$ coincide everywhere in that neighborhood. In the former case, the area increases faster on $\mathcal{M}$ than $\mathcal{N}$ when Σ is pushed forwards in time sufficiently close to the point p; in the latter case, the area increases are the same for $\mathcal{N}$ and $\mathcal{M}$ in the whole neighborhood.*

Proof. Consider the intersection of the two hypersurfaces and the partial Cauchy surface Σ. On Σ, let the shortest proper distance from $\mathcal{M}$ to $\mathcal{N}$ be f as a function on $\mathcal{M}$. The proper distance f is defined as the length of the spacelike geodesic connecting $\mathcal{N}$ and $\mathcal{M}$, and orthogonal to $\mathcal{M}$. Within a small neighborhood of a small size ϵ around g_0, the distance f can be expanded as a power series of ϵ. Since $\mathcal{N}$, $\mathcal{M}$ touch each other at g_0 and share the same tangent space there, the zeroth and first order terms of the expansion vanish, which implies that $f = \mathcal{O}(\epsilon^2)$. Hence, points on $\mathcal{N}$ and $\mathcal{M}$ are extremely close to each other in this neighborhood. Although we define f as a function on $\mathcal{M}$, we can also define a similar function on $\mathcal{N}$, but the difference between the two distances is negligible in this small neighborhood. Indeed, if we introduce a coordinate system on Σ, then these coordinates vary by a small amount of order f when we move from $\mathcal{N}$ to $\mathcal{M}$ and vice versa. Thus, we have $f(\mathcal{N}) = f(\mathcal{M}) + O(f\nabla f) = f(\mathcal{M}) + \mathcal{O}(\epsilon^3)$. Therefore, up to the order ϵ^2 or the order f, the domain in the definition of f is ambiguous, and it can be set to be either $\mathcal{N}$ or $\mathcal{M}$. In the small neighborhood of g_0, this also allows us to identify any two points on $\mathcal{N}$ and $\mathcal{M}$ which are connected by a geodesic defining f. As a result, this enables us to compare null generating vectors k^a on $\mathcal{N}$ and $\mathcal{M}$. When the normal vector k^a of $\mathcal{N}$ or $\mathcal{M}$ is projected onto Σ, it becomes $k^a|_\Sigma = cn^a$ where n^a is a normal unit vector pointing to the exterior of $\mathcal{N}$ or $\mathcal{M}$, and $c > 0$ is a constant. We can rescale the affine parameter of generators on $\mathcal{N}$ and $\mathcal{M}$ so that $c = 1$.

Now, consider a Gaussian normal coordinates (z, y^i) $(i = 1, \ldots, D-2)$ where y^i are Riemannian normal coordinates on $\mathcal{M} \cap \Sigma$ with g_0 as the origin. By using this coordinate system in a sufficiently small region around g_0 (up to order ϵ), the induced metric on Σ becomes a Euclidean metric and the Christoffel symbols vanish. By the construction, the surface $\mathcal{N} \cap \Sigma$ is given by the equation $z = f(y^i)$ where f is the proper distance as introduced above. The vectors normal to $\mathcal{M} \cap \Sigma$ and $\mathcal{N} \cap \Sigma$ are respectively given by

$$ {}^{(\mathcal{M})}N^\mu = (1, 0), \qquad {}^{(\mathcal{N})}N^\mu = (1, -\partial_i f). \tag{4.15} $$

Normalizing these vectors, we obtain

$$ {}^{(\mathcal{M})}n^\mu = (1, 0), \qquad {}^{(\mathcal{N})}n^\mu = \frac{(1, -\partial_i f)}{\sqrt{1 + (\partial_i f)^2}} = (1, -\partial_i f) + O\Big((\partial_i f)^2\Big) \tag{4.16} $$

$$ \Rightarrow \quad {}^{(\mathcal{M})}n^\mu \; - \; {}^{(\mathcal{N})}n^\mu = (0, \partial_i f) + O\Big((\partial_i f)^2\Big). \tag{4.17} $$

Therefore, the difference between normal vectors is

$$ {}^{(\mathcal{M})}n^a \; - \; {}^{(\mathcal{N})}n^a = \nabla^a f + O\Big((\nabla f)^2\Big). \tag{4.18} $$

In addition, $k^a = t^a + n^a$ where t^a is the timelike unit vector normal to Σ. Since t^a varies negligibly between $\mathcal{N}$ and $\mathcal{M}$ in this small neighborhood, we have

$$ {}^{(\mathcal{M})}k^a \; - \; {}^{(\mathcal{N})}k^a = \nabla^a f + O\Big((\nabla f)^2\Big) = \nabla^a f + \mathcal{O}(\epsilon^2). \tag{4.19} $$

Note that $\nabla^a f$ is tangent to $\mathcal{M} \cap \Sigma$ (or $\mathcal{N} \cap \Sigma$). Thus, the difference between the extrinsic curvatures of $\mathcal{M} \cap \Sigma$ and $\mathcal{N} \cap \Sigma$ is given by

$$ \hat{B}^{(\mathcal{M})}_{ab} - \hat{B}^{(\mathcal{N})}_{ab} = \nabla_a \nabla_b f. \tag{4.20} $$

Taking the trace with respect to the inverse metric h^{ab} pulled back to the null surface (either $\mathcal{N}$ or $\mathcal{M}$), we obtain

$$ \Delta\theta \equiv \theta^{(\mathcal{M})} - \theta^{(\mathcal{N})} = h^{ab} \nabla_a \nabla_b f \equiv \nabla^2 f. \tag{4.21} $$

Since the induced metric h_{ab} is Euclidean in the small neighborhood around g_0, let $r = \sqrt{\sum_i (y^i)^2}$ be the radial coordinate and $\mathrm{d}\sigma$ be the area element of a sphere of a radius $r = R$. To check the sign of $\Delta\theta$, we define a Green's function G on a small ball B of radius R centered at g_0 to be a solution of

$$ -\nabla^2 G(y) = \delta^{(D-2)}(y), \quad G|_{r=R} = 0. \tag{4.22} $$

The solution is given by

$$G(r) \propto \begin{cases} \left(\frac{1}{r^{D-4}} - \frac{1}{R^{D-4}}\right), & D > 4, \\ \ln\left(\frac{R}{r}\right), & D = 4, \end{cases} \tag{4.23}$$

where the proportional constant is positive. This shows that $G(r) > 0$ for any $r < R$, and hence $\partial_r G|_{r=R} < 0$. Then, we integrate $\Delta\theta$ over the ball B using G:

$$\begin{aligned} \int_B G\Delta\theta \mathrm{d}^{D-2}y = \int_B G\nabla^2 f \mathrm{d}^{D-2}y &= \int_{\partial B} G\boldsymbol{\nabla} f \cdot \mathrm{d}\boldsymbol{\sigma} - \int_B \boldsymbol{\nabla} G \cdot \boldsymbol{\nabla} f \mathrm{d}^{D-2}y \\ &= -\int_B \boldsymbol{\nabla} G \cdot \boldsymbol{\nabla} f \mathrm{d}^{D-2}y \\ &= -\int_{\partial B} f\boldsymbol{\nabla} G \cdot \mathrm{d}\boldsymbol{\sigma} + \int_B f\nabla^2 G \mathrm{d}^{D-2}y \\ &= -\int_{\partial B} f\partial_r G \,\mathrm{d}\sigma - f(g_0) \\ &= -\int_{\partial B} f\partial_r G \,\mathrm{d}\sigma \,. \end{aligned} \tag{4.24}$$

Since $\mathcal{M}$ never enters the exterior of $\mathcal{N}$, we have $f \geq 0$ everywhere, which implies that the integral must be non-negative. Using the fact that R can be arbitrarily small, if there exists a region around g_0 where $f > 0$, then the integral is strictly positive, which implies that there is at least one point in B where $\Delta\theta > 0$. Otherwise, if $f = 0$ everywhere in some neighborhood of g_0, then $\mathcal{N}$ and $\mathcal{M}$ coincide in that region. □

Having considered the Bekenstein-Hawking term, we move on to the entropy of matter in the exterior regions.

Lemma 4.6. *Suppose that the two hypersurfaces $\mathcal{N}$ and $\mathcal{M}$ coincide in a neighborhood of g_0, and Σ is evolved forwards in time to Σ' only in this neighborhood. That is, Σ and Σ' coincide outside the neighborhood of g_0 where $\mathcal{N}$ and $\mathcal{M}$ coincide. Then, the entropy S_{out} is increasing faster on $\mathcal{M}$ than on $\mathcal{N}$.*

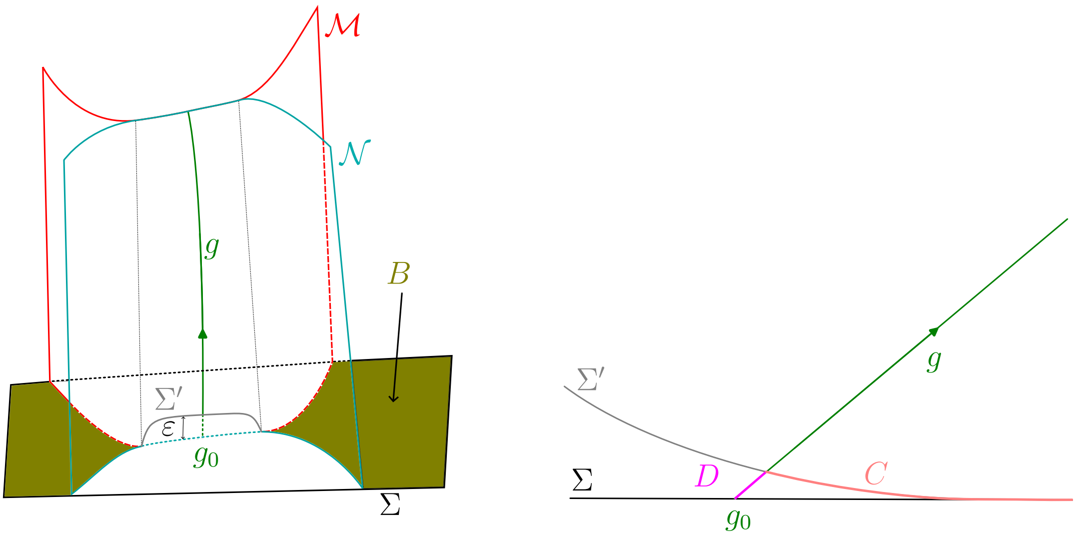

FIGURE 4.5: When the two surfaces $\mathcal{N}$ and $\mathcal{M}$ coincide in a neighborhood of g_0, we evolve Σ only within that neighborhood.

Proof. In quantum field theory, the von Neumann entropy is divergent due to the infinite degrees of freedom. However, the degree of entanglement between two disjoint systems B and C can be measured by a finite quantity called *mutual information* [28] defined by

$$I(B,C) = S(B) + S(C) - S(B \cup C), \tag{4.25}$$

which is always nonnegative because the von Neumann entropy is subadditive. Moreover, due to the strong subadditivity of von Neumann entropy, the mutual information increases when an extra system D is added [28], i.e.

$$I(B, C \cup D) \geq I(B,C) \quad \Rightarrow \quad I(B, C \cup D) - I(B,C) \geq 0. \tag{4.26}$$

Using (4.25), we obtain

$$S(B \cup C) - S(B \cup C \cup D) + S(C \cup D) - S(C) \geq 0. \tag{4.27}$$

To prove the statement, we define

$$B = \mathrm{Int}(\mathcal{N}) \cap \mathrm{Ext}(\mathcal{M}) \cap \Sigma = \mathrm{Int}(\mathcal{N}) \cap \mathrm{Ext}(\mathcal{M}) \cap \Sigma', \tag{4.28}$$

$$C = \mathrm{Ext}(\mathcal{N}) \cap \Sigma', \qquad D = \mathcal{M} \cap \Delta\Sigma = \mathcal{N} \cap \Delta\Sigma, \tag{4.29}$$

where $\Delta\Sigma = I^-(\Sigma') \cap I^+(\Sigma)$ is the spacetime region between Σ and Σ'. By definition,

$$B \cup C = \mathrm{Ext}(\mathcal{M}) \cap \Sigma'. \tag{4.30}$$

The region D contains all of matter that cross the surfaces $\mathcal{N}$ and $\mathcal{M}$ from the exterior to the interior when Σ evolves to Σ'. Since the evolution of Σ is within an infinitesimal time ϵ, D is approximately at the same time as Σ'. Therefore, $\mathrm{Ext}(\mathcal{N}) \cap \Sigma$ and $\mathrm{Ext}(\mathcal{M}) \cap \Sigma$ evolve into $C \cup D$ and $B \cup C \cup D$, respectively. Using fact that the fine-grained entropy is constant under unitary evolution, we obtain

$$S(C \cup D) = S(\mathrm{Ext}(\mathcal{N}) \cap \Sigma), \tag{4.31}$$
$$S(B \cup C \cup D) = S(\mathrm{Ext}(\mathcal{M}) \cap \Sigma). \tag{4.32}$$

As a result, we obtain

$$\Delta S_{\mathrm{out}}(\mathrm{Ext}(\mathcal{M}) \cap \Sigma) \geq \Delta S_{\mathrm{out}}(\mathrm{Ext}(\mathcal{N}) \cap \Sigma) \tag{4.33}$$

as desired. □

Proof of Theorem 4.4. Let a small neighborhood of g_0 on Σ be advanced to the future by an amount $\mathrm{d}\lambda$ of the affine parameter. Suppose that $\mathrm{d}S_{\mathrm{BH}}^{(\mathcal{M})} - \mathrm{d}S_{\mathrm{BH}}^{(\mathcal{N})} = \mathcal{O}(\hbar^p)$ and $\mathrm{d}S_{\mathrm{out}}^{(\mathcal{M})} - \mathrm{d}S_{\mathrm{out}}^{(\mathcal{N})} = \mathcal{O}(\hbar^q)$. Lemma 4.5 shows that $\mathrm{d}S_{\mathrm{BH}}^{(\mathcal{M})} - \mathrm{d}S_{\mathrm{BH}}^{(\mathcal{N})} > 0$ if p is finite. If $q > p$, then the Bekenstein-Hawking term dominates, which implies the statement. If $q < p$, then the von Neumann entropy term dominates. However, up to the order $\hbar^q$, $\mathcal{N}$ and $\mathcal{M}$ coincide on the small neighborhood around g_0, which implies that $\mathrm{d}S_{\mathrm{out}}^{(\mathcal{M})} - \mathrm{d}S_{\mathrm{out}}^{(\mathcal{N})} \geq 0$ by Lemma 4.6. □

4.2.2 Penrose-Wall singularity theorem

Using the GSL, we obtain a result similar to Lemma 3.2 which is obtained from NEC.

Theorem 4.7. *Assume that the GSL holds. Let g be a future-directed generator g of a null hypersurface $\mathcal{N}$ and let Σ be a partial Cauchy surface that intersects g at g_0. If the quantum expansion $\Theta^{(\mathcal{N})}[\Sigma; g_0]$ at g_0 is negative, i.e. the generalized entropy is decreasing in any small neighborhood of g_0, then g terminates in the future at some point on $\mathcal{N}$.*

Proof. Assume by contradiction that g is future-infinite. Let $\mathcal{M} = \partial I^-(g)$ be the causal horizon of a photon following g. Then, the two surfaces $\mathcal{M}$ and $\mathcal{N}$ touch each other at g. Since g is a subset of $\mathcal{N}$, the past set of g must be contained in the past set of $\mathcal{N}$. Therefore, we have

$$I^-(\mathcal{M}) \subset I^-(\mathcal{N}) \quad \Rightarrow \quad \mathrm{Ext}(\mathcal{M}) \subset \mathrm{Ext}(\mathcal{N}). \tag{4.34}$$

By the touching lemma, we have

$$\Theta^{(\mathcal{M})}[\Sigma; g_0] \leq \Theta^{(\mathcal{N})}[\Sigma; g_0] < 0,$$

which violates the GSL. □

Assume the spacetime (M, g_{ab}) is globally hyperbolic with a Cauchy surface Σ. On Σ, suppose there is a compact surface $\mathcal{T}$ of codimension 2 with a noncompact exterior. Moreover, assume that the inside and outside matter form a composite system in a pure state. Hence, the entropies of the inside and the outside are equal. The *ingoing* and *outgoing* light rays orthogonal to $\mathcal{T}$ form null hypersurfaces $\mathcal{N}_-$ and $\mathcal{N}_+$, respectively. Then, $\mathcal{T}$ is a *quantum trapped surface* if $\Theta^{(-)}[\Sigma; p] < 0$ and $\Theta^{(+)}[\Sigma; p] < 0$ for all $p \in \mathcal{T}$.

Theorem 4.8 (Wall, 2013)**.** *A spacetime (M, g_{ab}) is null geodesically incomplete if:*

(1) The GSL holds;

(2) There is a non-compact Cauchy surface where quantum matter is in a pure state;

(3) There is a quantum trapped surface $\mathcal{T}$.

Proof. Assume by contradiction that (M, g_{ab}) is null geodesically complete. Although the past sets of $\mathcal{N}_-$ and $\mathcal{N}_+$ are in the opposite sides, the assumption in (2) that the total state of matter is a pure state allows us to apply Theorem 4.7 to both $\mathcal{N}_-$ and $\mathcal{N}_+$. Therefore, $\partial I^+(\mathcal{T}) = \mathcal{N}_+ \cup \mathcal{N}_-$ is compact. As a result, we can apply the same argument in the proof of Theorem 3.6. □

Chapter 5

Summary and Recent Developments

5.1 Summary

- The Penrose singularity theorem relies on a classical energy condition (NEC) which can be violated by quantum effects.
- The GSL is a quantum condition that can replace NEC in the singularity theorem.
- The touching lemma is the key result that plays a role similar to the Raychaudhuri equation in the proof of Penrose theorem.
- When a quantum trapped surface is formed, a singularity or a Cauchy horizon must form in the semi-classical regime.

5.2 Recent Developments and Open Questions

- Wall's proof of the touching lemma relies on perturbative arguments. A non-perturbative proof is recently given by Shahbazi-Moghaddam (2025) [18] in holographic brane-world models.
- Recently, Bousso (2025) [26] proved a more robust singularity theorem that sidesteps the touching lemma by using *trapped spacetime wedge*.
- There are other quantum singularity theorems such as those by Bousso & Shahbazi-Moghaddam [29, 30] using the *Bousso bound*.

All of these developments are centered around Penrose's theorem (1965) which focuses on *NEC* and *null geodesic incompleteness*. On the other hand, Hawking's theorem (1967) [7] uses *SEC* (and some conditions) to prove *timelike geodesic incompleteness.*

Theorem 5.1 (Hawking, 1967)**.** (M, g_{ab}) *cannot be timelike and null geodesically complete if*

(1) $R_{ab}\zeta^a\zeta^b \geq 0$ *for every non-spacelike vector* ζ^a*;*

(2) the strong causality condition holds on (M, g_{ab})*;*

(3) there is some past-directed unit timelike vector ξ^a *at a point* p *and a positive constant* b *such that if* u^a *is the unit tangent vector to the past-directed timelike geodesics through* p*, then on each such geodesic the expansion* $\theta = \nabla_a u^a$ *of these geodesics becomes less than* $-3c/b$ *within a distance* b/c *from* p*, where* $c = -\xi^a u_a$*,*

Instead of global hyperbolicity, this theorem assumes a weaker condition, strong causality. One can wonder whether there exists a quantum generalization of this theorem.

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